Retrofitting Pretrained Transformers with Progressive Retrieval Attention: Bounded Native-K/V Memory for Qwen and Llama

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Abstract

Progressive Retrieval Attention (PRA) selects sparse, layer-native key/value (K/V) memory from references or displaced prompt history. Controlled models establish the mechanism but do not show that it can preserve a real pretrained transformer's attention semantics. We begin a pretrained integration study with a shared PRA execution core and thin Hugging Face attention adapters. The first adapter wraps one upper layer of the frozen 596M-parameter `Qwen/Qwen3-0.6B` checkpoint. When memory is disabled, the wrapper delegates to the original module and obtains bit-exact logits, hidden states, greedy generation, and all 28 generation-cache shapes. When memory is enabled, it reuses Qwen's pretrained projections, Q/K normalization, rotary embedding, grouped-query layout, causal mask, cache update, eager attention function, and output projection. Stored memory retains eight native K/V heads rather than expanding to 16 query heads. A CUDA smoke run materializes reference K/V from CPU and executes a 408-token logical prompt under a 256-token native-operation bound. A subsequent evidence-ranking study separates routing from materialization: post-RoPE key means, pre-RoPE key means, and attention-input hidden-state means all retain the same post-RoPE detail K/V. On 16 matched HotpotQA/QASPER examples with 32-token chunks, post-RoPE routing has a strong late- position correlation ($r = 0.652$) and recall@3 of 0.125. Pre-RoPE routing removes that correlation ($r = 0.009$), while hidden-state routing gives the best sparse recall@3, 0.438. Two confirmation subsets totaling 64 evaluations give hidden-state recall@3/8/16 of 0.391/0.578/0.797, but QASPER recall@3 is only 0.156 and recall@16 selects 23.8% of chunks. Splitting each 32-token parent into 2, 4, or 8 contiguous mean gists does not improve sparse ranking: recall@3 falls to 0.313/0.266/0.281 and MRR falls in every condition. A token-resolution maximum over 32 hidden states also fails as a general remedy: recall@3 changes from 0.438 to 0.375 while routing-index overhead grows from 1.57% to 50% of detail-K/V bytes. A final centered-RoPE control pools pre-RoPE keys and then rotates once at the exact span center. It improves recall@3 over a post-RoPE mean from 0.125 to 0.188, but remains below pre-RoPE (0.313) and hidden-state (0.438) routing. With eight centered subgists, correlation with native token-QK maximum ordering rises to 0.874 while recall@3 reaches only 0.250. Thus semantic selection and positional K/V transport should be separated, but parameter-free Qwen routing does not pass the predeclared quality gate. A held-out query study then rejects recent-state and question-span pooling: the last state obtains recall@3/MRR of 0.469/0.413, versus 0.250/0.279 for the strongest question-decay candidate. Evidence-supervised projections improve retrieval. Validation across capacity and objective controls selects an asymmetric 128-dimensional margin router with 262,144 parameters (0.044% of Qwen). Across five seeds it raises held-out recall@3 from 0.469 to 0.631 ± 0.069 (95% half-width), with HotpotQA/QASPER recall@3 of 0.588/0.675 and recall@8 of 0.806. Width 256, sampled negatives, and one hard-negative refresh do not improve this selection. Cross-domain recall@3 remains 0.213 in both directions and the 0.70 promotion gate is not met. A causal depth sweep then fixes oracle parent identities while replaying each identity's layer-native K/V. On four HotpotQA examples, mean gold-token log-probability rises by 1.004, 2.311, and 3.641 nats when the last 4, 8, and 14 layers consume memory, respectively, versus 4.635 for direct evidence text. All 28 layers are harmful (−0.766), and early or sparse placement is strongly harmful. QASPER becomes weakly positive only at 8 layers (+0.347), but its direct-text control is negative, so it remains non-diagnostic for transport quality. Routing once with the learned metric and replaying the selected IDs through the last 8 layers gives a combined +0.227 nats/token while preserving generated F1; the last-half setting gives +0.374 but lowers F1. The learned ranking reaches evidence-identity recall 0.264/0.403/0.521/0.639 at 5/10/20/30% of parent chunks. A bounded 416-token `#_head` probe retrieves the early fact at rank 1 and gains 5.469 nats/token through the last 8 layers, yet greedy decoding still misses it. Multi-layer consumption therefore establishes causal use of frozen native K/V, but not an end-task accuracy gain. We then productize the validated boundary as `pra-hf`: a stable Python API, CLI, fraction-based selection, Hugging Face-style router artifacts, and thin Qwen and Llama adapters. The Llama-family port uses the public 135M-parameter SmolLM2 checkpoint and passes exact disabled logit, hidden-state, generation-cache, and greedy-generation parity while preserving three native K/V heads of width 64. Five QASPER-trained seeds give QASPER evidence-identity recall@20% of 0.454 ± 0.007 for Qwen and 0.447 ± 0.051 for Llama-family routing. Four-example product-API probes cover annotated evidence in 0.667 and 0.750 of cases but obtain only 0.033 and 0.028 routed answer F1. Thus the release establishes a portable bounded-memory mechanism and measurable retrieval frontier, not solved long-context question answering.

1 Introduction

Pretrained decoder families differ in projection layout, rotary implementation, grouped-query attention (GQA), masks, sliding windows, and generation-cache conventions. Replacing a complete model class to add memory duplicates rapidly changing library code and makes parity difficult to audit. The alternative studied here is narrower:

$$\text{normal pretrained model} \rightarrow \text{wrap selected attention modules} \rightarrow \text{shared PRA core.}$$

The family adapter owns only native attention semantics. The shared core owns references, chunk gists, exact routing, the materialization budget, selected K/V transfer, and metrics. This boundary follows the broader transformer interface of queries attending to keys and values [9], while preserving Qwen's pretrained RoPE and GQA choices rather than approximating them in a parallel implementation [8, 1, 7].

This paper begins with Qwen because a sub-billion-parameter checkpoint makes exact tests practical on constrained hardware while still exercising modern RoPE and GQA. It then ports the same execution core to a Llama-family checkpoint through a family adapter that selects only the installed rotary and eager-attention functions. The contribution is a reproducible two-family reference implementation and an evidence hierarchy that prevents working transport or routing from being mistaken for useful answer generation.

2 PRA Recap and Prior Contracts

For layer ℓ, PRA stores reference chunks as native $K_c^{(\ell)}, V_c^{(\ell)} \in \mathbb{R}^{H_{kv} \times T_c \times d_h}$ and one or more compact gists $g_c^{(\ell)}$. A query representation $r^{(\ell)}$ scores the gists, and a global budgeter selects whole chunks. Materialized memory precedes the direct K/V sequence under one softmax:

$$\mathrm{Attn}(Q, [K_M; K_D], [V_M; V_D]) = \mathrm{softmax}\left(\frac{Q[K_M; K_D]^\top}{\sqrt{d_h}} + M\right)[V_M; V_D]. \tag{1}$$

This is not a separately normalized cross-attention residual. The native output projection is applied once to the combined result.

Paper 1 established controlled native-K/V routing and materialization. Paper 1.5 separated logical source position from contextual fragmentation. Continuous sources use $p_i = p_{\text{logical start}} + i$, and ordinary post-RoPE keys remain the canonical cache state. The present adapter preserves that contract by capturing each selected HF layer's actual post-RoPE keys and by assigning source-relative positions across bounded encoding blocks.

3 Adapter Architecture

Logical source / long prompt
↓ model-safe encoding blocks with logical offsets
↓ attention-input hidden state R + native post-RoPE K and native V
↓ smaller routing chunks → compact gists / packed index
↓ exact semantic routing → selected chunk identities
↓ whole-chunk materialization budgeter → selected native K/V
↓ direct K/V + memory K/V under one native attention softmax
↓ pretrained output projection

Figure 1: Canonical HF path. Routing uses compact semantic state, whereas attention receives the host model's selected native positional K/V payload. Model-family projection, RoPE, GQA, mask, cache, and output semantics remain in the thin adapter.

3.1 Routing State Is Not Attention State

PRA represents a cached chunk as

$$\mathcal{M}_c = (R_c, K_c, V_c), \tag{2}$$

where R_c is a compact routing representation and (K_c, V_c) is the payload used after selection. The objects need not inhabit the same feature space. The canonical HF configuration in this paper sets R_c to one or more gists of the hidden state entering the selected attention block, while K_c is that block's native post-RoPE key tensor and V_c its native value tensor. Token-level hidden states are transient during publication; only compact gists and detail K/V persist.

This split follows the specialization chain

$$h_{\text{in}} \longrightarrow W_K h_{\text{in}} \longrightarrow \text{RoPE}(W_K h_{\text{in}}). \tag{3}$$

The first representation retains broad contextual information, the second is an attention-address representation, and the third conditions that address on token position. Native token attention needs the third. Chunk retrieval may prefer the first. The routing query therefore uses the final attention-input hidden state and is compared only with hidden-state gists. It is never compared directly with Q_{post}. This is an empirical default for the tested Qwen checkpoint, not a claim that hidden states are universally optimal routers.

3.2 Shared execution core

The controlled model's former monolithic attention forward pass was divided into reusable operations. `prepare_pra_query` selects the newest valid query and maps GQA query groups into the native K/V routing width. `route_memory` calls the exact packed-index router. `budget_selected_memory` enforces

$$L_{\text{direct}} + L_{\text{materialized}} + L_{\text{reserve}} \leq L_{\text{native,max}}. \tag{4}$$

`materialize_selected_kv` removes duplicate overlap, preserves row isolation, and moves only retained detail K/V. `combine_local_and_memory_kv` pads variable rows and prepends an additive all-visible memory mask to the family's existing local causal mask. The controlled model and HF adapters now call this same core.

3.3 Thin HF contract

The abstract adapter exposes six family operations: project Q/K/V, apply native position encoding, normalize tensor layout, combine the native mask, invoke PRA attention, and project the output. Disabled memory takes an even narrower path: call the original attention object unchanged. This design preserves pretrained parameters and avoids checkpoint conversion.

The Qwen adapter has 272 physical lines (248 nonblank, non-comment lines). It introduces no learned parameter. Its family-specific branches distinguish Qwen2-style projections from Qwen3's Q/K normalization; routing, budgeting, materialization, residency, and metrics remain outside the file. This is a portability measurement, not a code-minimization target.

4 Qwen Native Semantics

4.1 Projection and RoPE

The adapter calls the wrapped module's `q_proj`, `k_proj`, and `v_proj`. Qwen3's existing `q_norm` and `k_norm` are applied before the installed Transformers `apply_rotary_pos_emb` function. Reference capture saves post-RoPE K and position-independent V. No independent RoPE implementation is introduced.

4.2 Grouped-query attention

The tested checkpoint has $H_q = 16$, $H_{kv} = 8$, and $d_h = 128$. Cache objects retain $[B, 8, T, 128]$ K/V. Key-space routing baselines average each pair of query heads sharing one K/V head, producing a vector in the same 8×128 space as a flattened key gist. The canonical hidden-state router instead uses one d_{model} vector and does not alter payload head layout. K/V expansion occurs inside Qwen's eager attention function, never in persistent PRA storage. A synthetic replay test compares this path with explicit K/V head repetition and matches within 10^{-6}.

4.3 HF cache and masks

Prefill and single-token decode both use the wrapped module's HF `Cache.update` call. PRA memory is prepended after that update, so direct generation history appears once. A decode test with four prompt tokens and two generated tokens observes five direct cache positions at the last invocation, plus four separate memory positions. The original causal mask remains on local/cache positions; selected historical memory receives a visible prefix mask. Memory- disabled generation delegates entirely to the original module.

5 Experimental Setup

Table 1 records the first checkpoint exactly. Tests use synthetic Qwen3 configs offline. The pre-trained run uses Python 3.10.11, PyTorch 2.12.1+CUDA 12.6, Transformers 4.55.4, eager attention, FP16, and an NVIDIA GeForce GTX 950M. The frozen model is not fine-tuned.

Table 1: Pinned Qwen checkpoint and PRA placement.

Checkpoint	`Qwen/Qwen3-0.6B`
Revision	`c1899de289a04d12100db370d81485cdf75e47ca`
Parameters / layers	596,049,920 / 28
Query heads / K/V heads / head width	16 / 8 / 128
RoPE base / advertised positions	10^6 / 40,960
Initial PRA placement	layer 27 only
Depth-study instrumentation / selected band	all 28 layers / layers 20–27
Initial gist / routing policy	mean, one gist/chunk, exact top-k
Native limit / direct / selected-memory cap	256 / 128 / 64 tokens
K/V residency	CPU full cache, selected transfer

Every run writes JSON and scalar CSV with the Git SHA, model revision, software versions, device, dtype, layer selection, limits, routing settings, timings, and metrics. The exact executed commit is retained in each artifact rather than inferred from the manuscript build.

6 Adapter Parity

PRA-disabled parity is the hard gate. The baseline is evaluated first; the same in-memory model is then wrapped, leaving every parameter object in place. Table 2 reports exact results for the five-token prompt "The capital of Portugal is" and four-token greedy decode.

Table 2: Original HF model versus memory-disabled PRA wrapper.

Metric	Result
Maximum / mean absolute logit difference	0 / 0
Maximum hidden-state difference	0
Greedy token agreement	100%
Greedy sequences exactly equal	yes
Generation-cache shapes equal	all 28 layers
Finite adapted logits	yes

Single-run prefill times were 0.447 s before wrapping and 0.136 s after wrapping. This order is confounded by first-call warm-up and is not a speed comparison. Exact outputs, rather than latency, are the parity evidence.

7 Native-KV Reference Smoke

An explicit 22-token reference about Lisbon is encoded through the frozen model. The selected layer captures $[1, 8, 22, 128]$ K/V, stores it on CPU, creates a mean gist, and transfers all 22 tokens because the source fits one routing chunk. The physical transfer is 90,112 bytes, exactly $2 \times 22 \times 8 \times 128 \times 2$ bytes for FP16 K and V. The run reports 1.6 ms routing, 0.40 ms materialization orchestration, 0.15 ms selected transfer, and 0.11 ms combined attention. These synchronized phase values are instrumentation checks on one machine, not throughput claims. Cold reference construction took 0.144 s and the warm query took 0.191 s.

The generated continuation contains "Lisbon", but the unmodified model also knows this fact. Consequently, this observation establishes functional selected-K/V transport and generation, not a causal retrieval benefit. The exact native-replay claim is instead restricted to the synthetic tensor test, where GQA memory plus local K/V equals explicit dense composition.

8 Implicit Head and Bounded Long Prompt

A repeated prompt tokenizes to 408 logical tokens. The direct cap retains the final 128 tokens; the first 280 become `#__head`. That head is encoded in blocks of at most 64 tokens, with source-relative positions. The direct tail receives positions 280 through 407. Every encoding operation and the final selected-memory operation stays under the 256-token configured native limit; the largest observed source-encoding call is 64 tokens and violations are zero. The final finite-logit query took 0.258 s. This proves bounded execution and offset propagation for the frozen Qwen adapter. It does not show that the router selected the information needed by an arbitrary continuation.

9 Unrestricted QA Smoke

We next run one fixed HotpotQA example [10] and one QASPER example [3]. Four conditions use the same frozen checkpoint: question-only truncation, dense text left-truncated to 256 tokens, oracle evidence text-RAG, and PRA with the question direct and the complete source in native-K/V memory. Oracle text-RAG is an upper control because it receives annotated evidence. This two-example matrix diagnoses plumbing; it cannot estimate dataset accuracy.

Table 3: Unrestricted-QA smoke outcomes. Standard normalized token F1 is shown.

Dataset	Condition	F1	Annotated evidence selected
HotpotQA	truncation; dense; oracle text-RAG; PRA	0; 0; 0; 0	PRA: 0%
QASPER	truncation; dense; oracle text-RAG; PRA	0; 0.182; 0; 0.182	PRA: 0%

HotpotQA's expected answer is "yes"; every condition begins with "No". QASPER's expected answer is "no"; dense and PRA generations begin with "No", while truncation begins with "Yes". However, PRA routes three tail chunks, none overlapping the annotated evidence, so the QASPER output cannot be credited to retrieved evidence. Qwen also fails the oracle text-RAG control on both examples. The proper conclusion is negative: this frozen top-layer, one-gist-per-chunk configuration has not demonstrated content-causal retrieval.

The selections expose a concrete failure mode. For HotpotQA, evidence lies at token spans 107–137 and 543–588, while routed chunks lie near 1216–1318. For QASPER, evidence lies at 0–317, while selected chunks lie near 4032–4192. Post-RoPE mean-key gists and an unadapted top-layer query systematically prefer irrelevant late-source chunks in these two probes.

10 Routing Representation Study

The smoke failure motivates a narrower test: is the post-RoPE representation appropriate for semantic routing? For token j at position p_j, the baseline pools

$$g_{\text{post}} = \frac{1}{m} \sum_{j=1}^{m} R(p_j) K_{\text{pre},j}, \tag{5}$$

which mixes content with different rotary phases. The position-neutral alternative pools $g_{\text{pre}} = m^{-1} \sum_j K_{\text{pre},j}$. A third representation pools the attention-input hidden states. These are hypotheses about ranking only; selected detail memory remains Qwen's post-RoPE K and native V in every condition.

attention-input hidden states

$\swarrow$ hidden-state mean gist | native Q/K/V projection and Q/K norm $\searrow$

semantic routing $\leftarrow$ pre-RoPE Q/K $\xrightarrow{\text{RoPE}}$ post-RoPE Q/K

$\uparrow$ pre-RoPE key gist | $\downarrow$ post-RoPE detail K + native V

exact selection $\longrightarrow$ selected native K/V $\longrightarrow$ Qwen attention

Figure 2: Routing and transport are separate. Pre-RoPE or hidden-state features rank chunks; the attention path always materializes post-RoPE native K/V.

The code makes this independence executable. Routing returns stable *selected chunk identities*; a separate core entry point applies the common budget and materializes their payload. A fixed-selection test supplies the same identity set with ordinary-router and oracle metadata. It obtains bit-identical post-RoPE K, identical V, and exactly identical native attention output. Router choice can change the set, but it cannot silently change what a fixed set means to attention.

10.1 Protocol

Stage 1 uses eight HotpotQA and eight QASPER validation examples, matched across representations. The frozen checkpoint, last-layer placement, one mean gist per chunk, 128-token encoding blocks, zero routing overlap, and exact tensorized search remain fixed. We compare post-RoPE keys, normalized pre-RoPE keys, and attention-input hidden states with 32-token routing chunks and $k \in \{3, 8, 16\}$. Evidence text is mapped to exact tokenizer spans. The final-token question query is captured at the same layer: GQA query groups are averaged into native K/V width for both key representations, while hidden-state routing uses the matched final hidden vector.

The experiment scores all chunks and does not generate answers. This keeps evidence ranking separate from decoding and from the hard materialization budget. Stage 2 carries the best two representations to 64- and 128-token chunks. Stage 3 confirms the best sparse configuration on two deterministic 32-example subsets with different seeds: 64 evaluations covering 60 unique examples. No model or router parameter is trained.

10.2 Representation and position bias

Table 4 reports Stage 1 means over both datasets. The selected fractions at $k = 3, 8, 16$ are 0.040, 0.107, and 0.213 for every representation. Post-RoPE scores are strongly correlated with later source position. Both position-neutral alternatives remove that correlation and improve evidence ranking.

Table 4: Stage 1 routing representation comparison, 16 matched examples and 32-token chunks.

Routing representation	Recall@3	Recall@8	Recall@16	MRR	Score-position r
Post-RoPE key mean	0.125	0.250	0.313	0.131	0.652
Pre-RoPE key mean	0.313	**0.563**	**0.750**	**0.301**	0.009
Hidden-state mean	**0.438**	0.500	**0.750**	0.276	-0.021

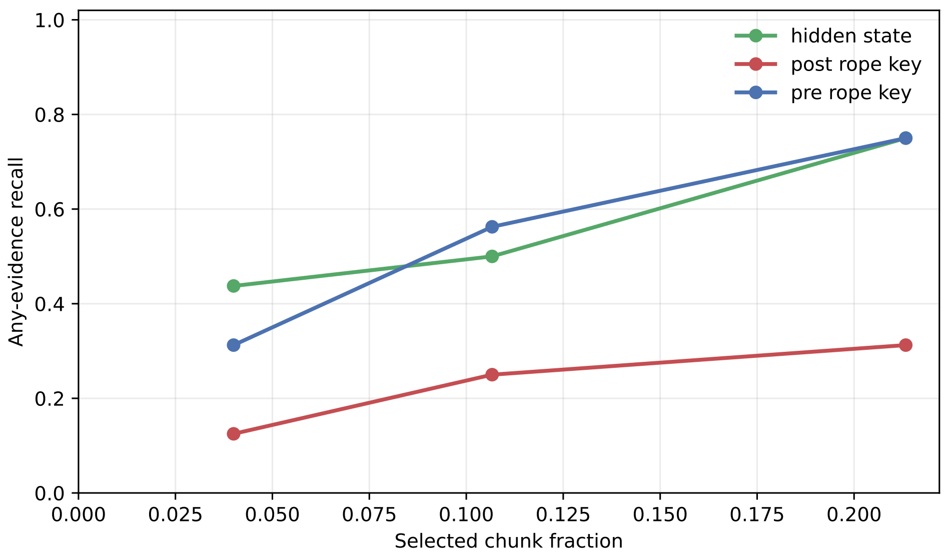

Figure 3: Any-evidence recall against routed chunk fraction in Stage 1. Pre-RoPE and hidden-state routing dominate the post-RoPE baseline, but high recall still requires broader selection.

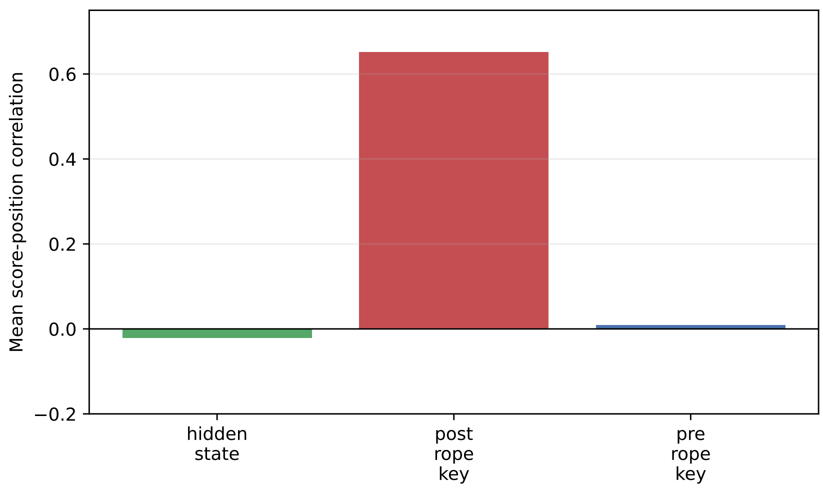

Figure 4: Mean chunk-score correlation with normalized source position. The late-source effect is specific to post-RoPE mean-key routing in this comparison.

The result supports the RoPE-contamination hypothesis: the post-RoPE representation is a poor mean-pooled semantic index. It does not establish that pre-RoPE keys are the best router. Hidden-state means provide the best low-k recall, while pre-RoPE keys provide the best MRR and recall@8. Native key space and semantic retrieval space need not coincide even before rotation.

10.3 Chunk size and breadth

Table 5 shows that larger chunks sometimes raise recall, but they also make a fixed k cover much more of the source. At 128 tokens and $k = 8$, both representations reach 0.688 recall while selecting 41.3% of chunks. At $k = 16$, the selected fraction is 71.5%. These are useful boundary measurements, not sparse-routing wins.

Table 5: Stage 2 chunk-size sweep over the 16 matched examples. Fraction is reported at $k = 8$.

Representation	Chunk tokens	Recall@3	Recall@8	Recall@16	Selected fraction@8
Pre-RoPE key	32	0.313	0.563	0.750	0.107
Pre-RoPE key	64	0.250	0.438	0.813	0.211
Pre-RoPE key	128	0.375	0.688	0.875	0.413
Hidden state	32	0.438	0.500	0.750	0.107
Hidden state	64	0.313	0.625	0.813	0.211
Hidden state	128	0.438	0.688	0.813	0.413

The expected larger-k recall/cost tradeoff is confirmed. The predicted degradation of post-RoPE pooling with larger chunks is not tested after that representation failed Stage 1; the two finalists do not degrade monotonically. Their larger-chunk gains are inseparable from coarser evidence coverage and a larger selected source fraction.

10.4 Independent confirmation and gate

Hidden-state routing with 32-token chunks is selected for confirmation because it gives the best recall at $k = 3$. Table 6 combines both seeds. Materialization is capped at 128 tokens, so routed breadth continues to grow at $k = 8$ and 16 while the physically materialized fraction stays near 0.059.

Table 6: Hidden-state confirmation over 64 evaluations. Intervals are 95% Wilson intervals.

k	Any recall	95% interval	All recall	Selected fraction	Materialized fraction
3	0.391	[0.281, 0.513]	0.063	0.045	0.045
8	0.578	[0.456, 0.691]	0.203	0.119	0.059
16	0.797	[0.683, 0.877]	0.500	0.238	0.059

At $k = 3$, HotpotQA recall is 0.625 but QASPER recall is 0.156; combined MRR is 0.326. The combined score-position correlation is -0.077, so the original late-position failure is not the remaining blocker. The predeclared promotion gate requires combined recall@3 ≥ 0.70, each dataset ≥ 0.60, MRR ≥ 0.50, selected fraction ≤ 0.10, and absolute position correlation ≤ 0.15. Only the last two conditions pass. Under the original sequential quality protocol, Qwen therefore does not promote a router to Llama. Section 13 later evaluates Llama as an independent software-portability gate and does not reinterpret this failed quality gate as a pass.

One hidden-state gist adds 1.57% over detail K/V bytes at 32-token chunks. On the GTX 950M, mean packed-index construction is 3.50 ms, warm exact routing plus selection is 1.90–2.74 ms, and the isolated `torch.topk` primitive is about 0.05 ms. These are single-request Python measurements, not serving-throughput claims.

10.5 Contiguous multi-gist diagnostic

Before introducing learned parameters, we test whether one arithmetic mean simply erases useful local structure. Let the $T = 32$ hidden states in parent chunk c be divided into G balanced, contiguous, nonempty subspans $S_{c,i}$. The routing representation and parent score are

$$g_{c,i} = \frac{1}{|S_{c,i}|} \sum_{t \in S_{c,i}} h_t, \qquad s_c = \max_{i \leq G} \cos(q, g_{c,i}). \tag{6}$$

For $G = 1, 2, 4, 8$, nominal subspan widths are 32, 16, 8, and 4 tokens. Top-k selection remains over parent chunks, not gists. A winning subspan therefore identifies one parent, whose post-RoPE native K/V is materialized once under the same 128-token budget. This isolates routing-index resolution from K/V storage and materialization granularity.

The implementation was fixed before the run at repository commit `0205df8`. Balanced pooling and local-span metadata occupy `gists/segment_mean.py`, lines 11–56. Reference publication calls the common gist interface at `hf/injection.py`, lines 163–190. The existing packed router stores $[C, G, D]$ gist tensors, masks short final chunks, reduces with a maximum, and retains the winning gist index at `memory.py`, lines 487–593. A unit gate requires $G = 1$ to be bit-exact with the historical single mean.

The exploratory sweep uses the same 16 matched examples as Stage 1. Since every multi-gist condition loses at recall@3, no intervention qualifies as a candidate under the predeclared selection rule. We nevertheless run the three interventions on both established 32-example subsets to test whether the negative direction persists. Table 7 combines 64 evaluations per G; four examples occur in both subsets, leaving 60 unique examples.

Table 7: Contiguous hidden-state means with fixed 32-token parents. Overhead is routing-gist bytes divided by post-RoPE detail K/V bytes.

G	Tokens/gist	Recall@3	95% interval	MRR	Recall@8/16	Overhead
1	32	**0.391**	[0.281, 0.513]	**0.326**	0.578/0.797	1.57%
2	16	0.313	[0.212, 0.434]	0.292	0.516/0.734	3.14%
4	8	0.266	[0.173, 0.385]	0.234	0.500/0.688	6.28%
8	4	0.281	[0.186, 0.401]	0.249	0.563/**0.828**	12.56%

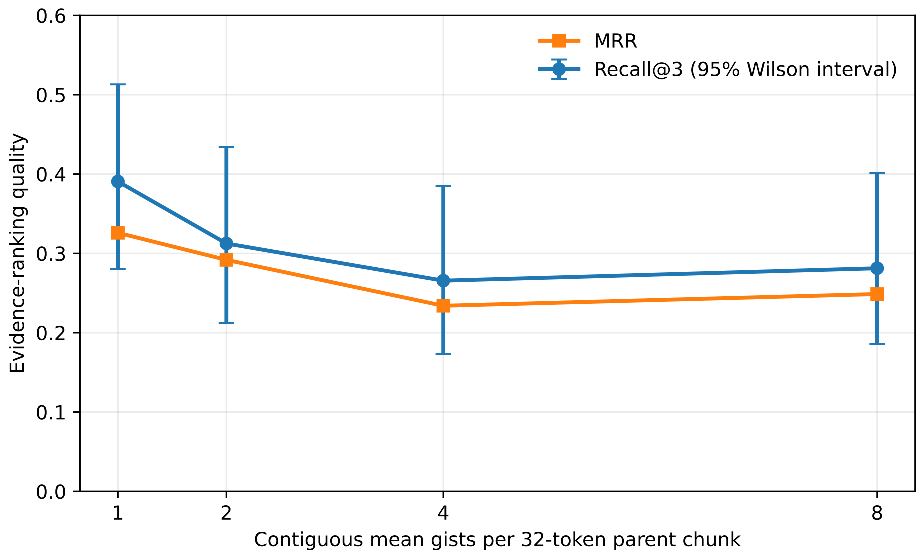

Figure 5: Sparse evidence ranking does not improve as a parent chunk receives more contiguous mean gists. Intervals are Wilson intervals over 64 evaluations.

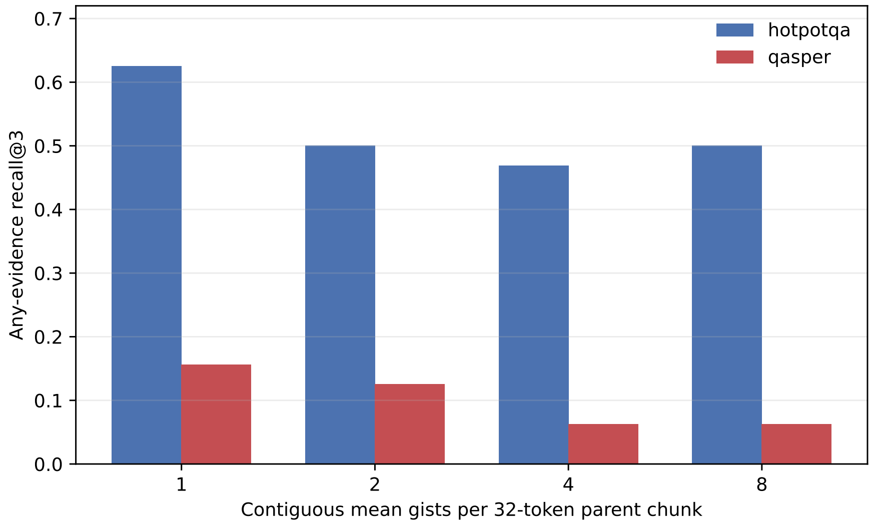

Figure 6: Recall@3 declines on both datasets. The QASPER weakness is not repaired by finer parameter-free means.

At $k = 3$, paired recall changes relative to $G = 1$ are 4 gains versus 9 losses for $G = 2$, 5 versus 13 for $G = 4$, and 5 versus 12 for $G = 8$; exact two-sided McNemar p-values are 0.267, 0.096, and 0.143. The intervals overlap and the paired tests do not cross 0.05, so this is not evidence that all multi-gist schemes are harmful. It is evidence against the specific proposed remedy: untrained contiguous means with max aggregation do not improve sparse ranking here. The $G = 8$ recall@16 increase is a broad-retrieval effect, not a sparse win; recall@3 and MRR remain below the single-mean baseline.

Routing-index bytes scale nearly linearly with G, while selected/materialized parent fractions and active K/V bytes remain identical at fixed k. Packed-index construction is noisy (3.50, 11.60, 4.62, and 4.81 ms for $G = 1, 2, 4, 8$), and warm routing remains 1.93–2.79 ms at this small scale; neither timing sequence supports a throughput claim. The result rejects H5, leaves the proposed H6 saturation inapplicable, and supports H7 for bytes and K/V materialization. It makes an evidence-supervised frozen-Qwen router the next controlled intervention before broader clustering, query-conditioned gists, or another model family.

10.6 Token-resolution hidden-state diagnostic

One remaining parameter-free question is whether useful evidence survives in individual hidden states even when every tested mean loses it. With the parent width fixed at 32 tokens, we set $G = 32$ in the same balanced segment implementation. Each nonempty segment is then one token, so the parent score becomes

$$s_c = \max_{t \in c} \cos(h_{\text{query}}, h_t). \tag{7}$$

This is an upper-resolution diagnostic, not a proposed production index. It preserves parent identity, selected fraction, post-RoPE K/V, and the materialization budget, while increasing the number of routing vectors by about 32 times.

Table 8: Matched 16-example hidden-state diagnostic with fixed 32-token parents.

Router	Recall@3	Recall@8	Recall@16	MRR	Index/detail-KV %
One parent mean ($G = 1$)	**0.438**	0.500	0.750	0.275	1.57
Maximum over tokens ($G = 32$)	0.375	**0.625**	**0.813**	**0.314**	50.00

The aggregate MRR gain is not consistent across datasets. HotpotQA improves from 0.338 to 0.469, with unchanged recall@3 of 0.625; QASPER MRR falls from 0.213 to 0.160 and recall@3 falls from 0.250 to 0.125. Figure 7 shows the corresponding breadth tradeoff. The result does not support token-max as a general sparse-routing remedy. It does suggest that pooling loss can matter for some HotpotQA ranks, while QASPER's dominant failure is weak query/evidence alignment even at token resolution.

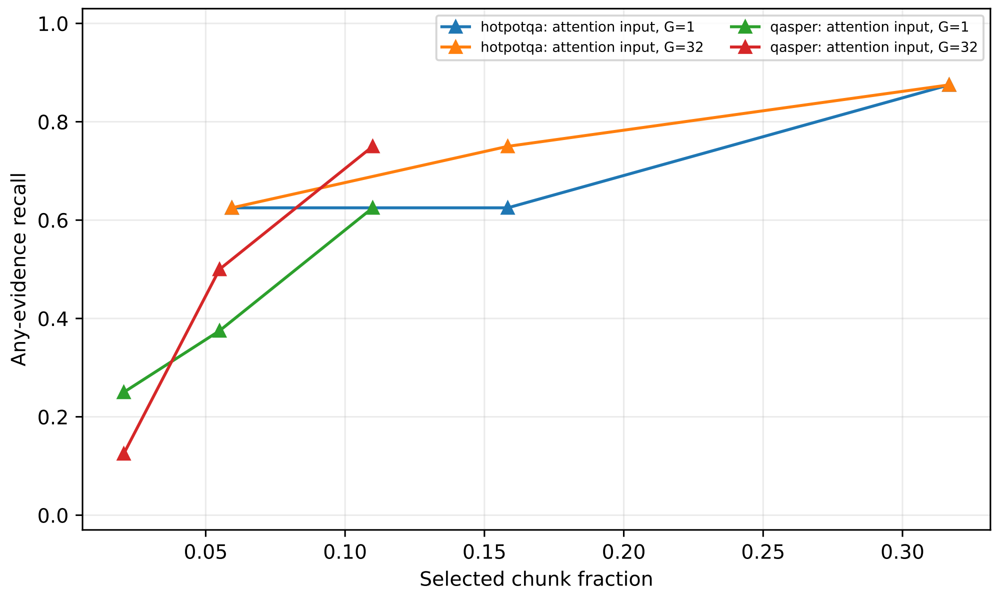

Figure 7: Token-resolution maxima improve some broader ranks but reduce sparse QASPER recall. Selected parent fractions, not routing-vector fractions, are shown.

The token-max index contains a mean 3,255 candidate gists per example versus 102 for one mean. Its measured index-construction time is about 40.1 ms versus 16.4 ms on this small run, while warm top-k timing is too noisy to interpret. The decisive cost is persistent routing state: 50% of detail-K/V bytes is incompatible with the intended compact-index role.

10.7 Center-Position RoPE Gists

The final zero-parameter control asks whether post-RoPE routing failed because it averaged keys expressed in different positional frames. For a contiguous span S_c with known source positions

$p_s, \ldots, p_e$, we distinguish

$$g_{\text{pre}} = \frac{1}{|S_c|} \sum_{t \in S_c} K_{\text{pre},t}, \tag{8}$$

$$g_{\text{postmean}} = \frac{1}{|S_c|} \sum_{t \in S_c} R(p_t) K_{\text{pre},t}, \tag{9}$$

$$g_{\text{center}} = R(p_c) g_{\text{pre}}, \qquad p_c = \frac{p_s + p_e}{2}. \tag{10}$$

The centered form pools in one position-neutral frame and introduces a single representative position afterward. Its matched routing query is $Q_{\text{post}} = R(q)Q_{\text{pre}}$. Even-length spans use their exact half-integer center through Qwen's installed rotary module; no independent RoPE approximation is introduced. This changes routing gists only. Every selected parent still materializes the same native post-RoPE K and native V.

The implementation is revision-pinned at commit `4c2a0f0`. Configuration and policy validation are in `hf/config.py`, lines 11–75. Qwen reshapes each pooled $[G, H_{kv} d_h]$ gist to native heads, asks the installed rotary module for float-position cosines and sines, applies the family helper, and restores packed routing layout at `hf/qwen.py`, lines 80–105. Span centers and persistent source-span metadata are constructed independently of packed-index order at `hf/injection.py`, lines 105–148; publication invokes that transform only after ordinary pre-RoPE pooling at lines 207–229.

Table 9 reports the same 16 examples, 32-token parents, and $k \in \{3, 8, 16\}$ used in the representation study. Centered-RoPE is directionally better than the post-RoPE mean, with one paired Recall@3 gain and no losses, but the exact two-sided McNemar p-value is 1.0 because only one example changes. Its position correlation falls from 0.651 to 0.567 but remains far above the position-neutral alternatives.

Table 9: Matched centered-RoPE representation control with one gist per 32-token parent.

Router	Recall@3	Recall@8	Recall@16	MRR	Position r
Post-RoPE mean	0.125	0.250	0.313	0.131	0.651
Centered-RoPE mean	0.188	0.313	0.438	0.168	0.567
Pre-RoPE mean	0.313	**0.563**	**0.750**	**0.301**	0.008
Hidden-state mean	**0.438**	0.500	**0.750**	0.275	-0.021

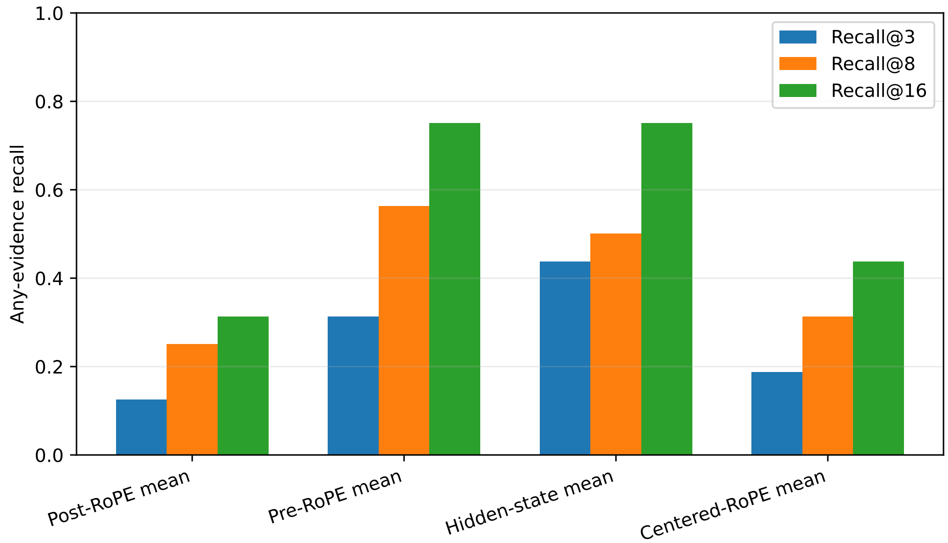

Figure 8: Centered pooling repairs part of the post-RoPE mean failure, but position-neutral pre-RoPE and hidden-state routing retain substantially better sparse evidence recall.

For segment means, each subspan receives its own exact center. At $G = 1, 2, 4, 8$, nominal spans contain 32, 16, 8, and 4 tokens, with maximum center-approximation errors of 15.5, 7.5, 3.5, and 1.5 positions. Table 10 compares evidence ranking with Spearman correlation against the chunk ordering produced by maximum native token-QK score.

Table 10: Centered segment gists. QK rank is Spearman correlation with native token-QK maximum ordering; overhead is routing-index bytes divided by detail-K/V bytes.

G	Recall@3	MRR	Position r	Native-max QK rank	Overhead
1	0.188	0.168	0.567	0.521	1.57%
2	0.125	0.134	0.588	0.602	3.14%
4	**0.250**	0.139	0.557	0.710	6.28%
8	**0.250**	**0.180**	0.455	**0.874**	12.56%

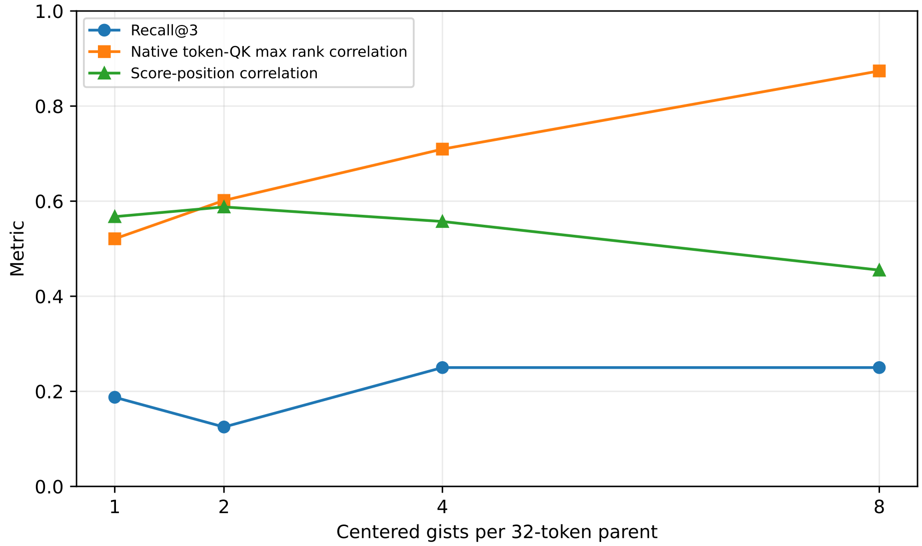

Figure 9: Finer centered gists increasingly reproduce native token-QK maximum ordering, but semantic Recall@3 remains weak. Native attention compatibility and evidence relevance are different objectives.

The mechanistic hypothesis is therefore supported, but the routing-quality hypothesis is not. At $G = 8$, native-maximum rank correlation reaches 0.874 and position bias declines, yet Recall@3 is 0.250 and QASPER Recall@3 remains zero. Exact, floor, and ceil center policies all produce Recall@3 of 0.125 on the matched eight-example fractional control; their score changes are below 0.0074, so fractional-position handling is not the blocker. E1 is directionally supported, E2 is rejected, E3 is supported, and E4 receives only weak partial support. This closes the current zero-parameter routing transformations and advances the frozen-Qwen learned hidden-state router.

10.8 Representing the Information Need

The preceding experiments vary the memory representation while using the final attention-input state as the retrieval query. We next hold memory fixed at one hidden-state mean per 32-token chunk and ask whether the information need should aggregate several prompt states. For prompt states $h_1, \ldots, h_t$, the recent exponential candidate is

$$r_q = \frac{\sum_{i=0}^{W-1} 2^{-i/H} h_{t-i}}{\sum_{i=0}^{W-1} 2^{-i/H}}, \tag{11}$$

where H is a token half-life. Exact-question variants apply the same weighting only to states whose tokenizer offsets overlap the known benchmark question. Question boundaries are evaluation metadata, not an assumption about ordinary generation.

The broad validation stage compares last state; uniform windows 4, 8, and 16; exponential $W = 16$ with $H = 2, 4, 8$; a linear control; question mean; and question decay $H = 4, 8$ on 8 examples per dataset. Last-state recall@3 is 0.438. Question decay at $H = 4$ is the strongest aggregate at 0.375 and raises HotpotQA MRR from 0.338 to 0.526, but lowers QASPER recall@3 from 0.250 to 0.125. Refinement adds half-lives 2 and 16 and windows 2 and 32. We carry the best question-decay and long-uniform tradeoffs to 32 identity-disjoint confirmation examples.

Table 11: Held-out query-representation confirmation with fixed hidden-state memory gists.

Query	Recall@3	Recall@8	Recall@16	MRR	Position r
Last state	**0.469**	**0.719**	**0.844**	**0.413**	-0.075
Question decay, $H = 2$	0.250	0.688	0.813	0.279	-0.014
Uniform suffix, $W = 32$	0.313	0.469	0.813	0.348	-0.076

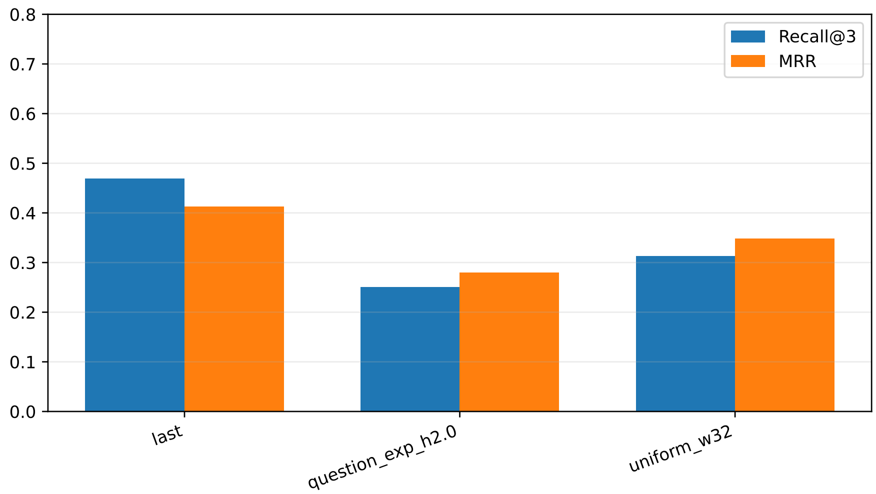

Figure 10: Neither selected aggregate reproduces the last-state sparse ranking on held-out questions. Aggregation changes query geometry substantially but does not improve relevance.

The aggregate vectors are not trivial perturbations: their mean cosine with the last state is 0.459 for question decay and 0.576 for the 32-token mean. Nonetheless, both lower sparse recall and MRR. A post-hoc length split suggests that the long uniform window is less harmful on long questions, but the bins contain only 21 and 11 examples and confound dataset composition. We do not promote a length-conditioned policy. The simplest last-state query remains canonical.

10.9 Learning Relevance Geometry on Frozen Hidden States

The negative resolution and query-pooling results motivate a learned metric rather than another hand-designed representation. We precompute frozen last-state queries and mean chunk gists for 48 train, 16 validation, and 32 test examples, split by QA identity. The sets contain 4,707, 1,634, and 3,084 candidate chunks, with no identity overlap. All 596,049,920 Qwen parameters have `requires_grad=False`. For positive chunks P_q among all in-document candidates C_q, the first adapters minimize

$$\mathcal{L}_q = -\log \frac{\sum_{c \in P_q} \exp(s(q, c)/\tau)}{\sum_{c \in C_q} \exp(s(q, c)/\tau)}. \tag{12}$$

The objective extension also evaluates pairwise margin ranking,

$$\mathcal{L}_q^{\text{margin}} = \frac{1}{|P_q||N_q|} \sum_{p \in P_q} \sum_{n \in N_q} \max(0, m - s(q, p) + s(q, n)), \tag{13}$$

with $m = 0.2$. Exhaustive candidates include current-router false positives, lexical competitors, and position-matched negatives. Separate controls sample seven mixed negatives per positive and then refresh that pool once with top false positives from the learned router.

We compare shared linear, asymmetric linear, and asymmetric two-layer MLP projections at widths 64 and 128 with five seeds, then add width 256 and the margin objective without opening the test set for selection. Validation recall@3 followed by MRR selects asymmetric linear width 128 with the margin objective (validation recall@3/MRR 0.838/0.692). It contains 262,144 parameters, 1.0 MiB in float32, or 0.044% of Qwen. Table 12 reports the held-out result.

Table 12: Five-seed held-out learned-routing controls. The confidence interval is the 95% half-width across adapter seeds. Selection uses validation only.

Metric	Recall@3	Recall@8	MRR	Hotpot R@3	QASPER R@3
Cosine, last state	0.469	0.719	0.413	0.813	0.125
Asymmetric-128, contrastive	0.563 ± 0.039	0.806	0.516	0.400	0.725
Shared-256, contrastive	0.631 ± 0.058	0.819	0.558	0.500	0.763
Asymmetric-128, margin (selected)	$\mathbf{0.631 \pm 0.069}$	0.806	0.498	0.588	0.675
Asymmetric-128, mixed negatives	0.538 ± 0.069	0.819	0.494	0.313	0.763
Asymmetric-128, mined round 1	0.506 ± 0.042	0.825	0.479	0.288	0.725

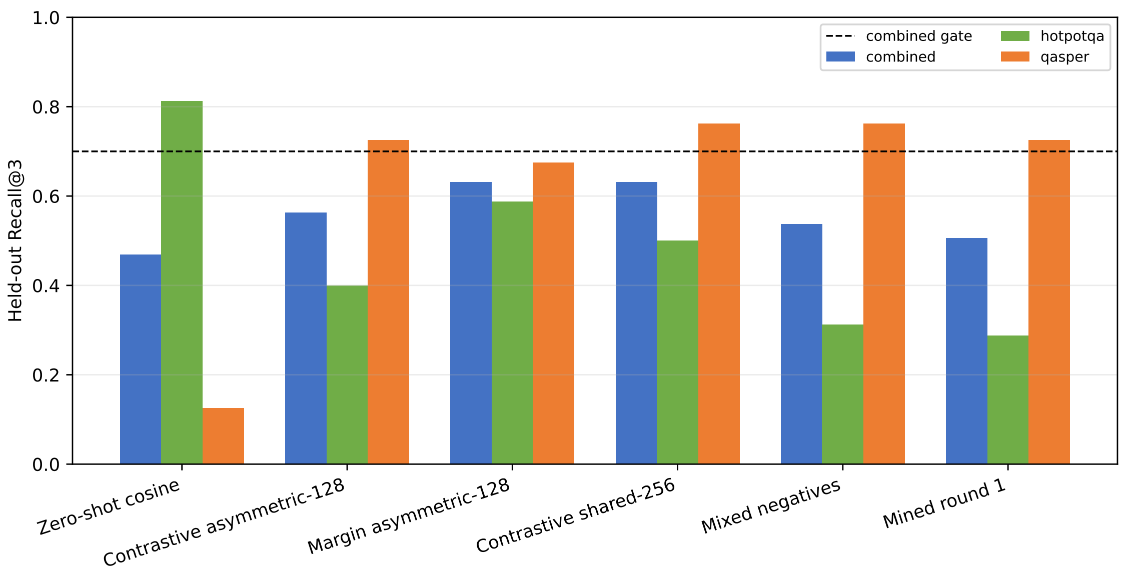

Figure 11: The margin objective balances HotpotQA and QASPER better than the contrastive adapter. Larger width and harder sampled negatives do not cross the predeclared combined gate.

The selected adapter improves recall@3 over zero-shot cosine by 0.163 ± 0.069 across the five paired seeds. It passes recall@8 (0.806), both per-dataset recall@3 thresholds, position bias ($r = -0.026$), and width gates, but fails combined recall@3 (0.631 versus 0.70). Shuffled-label recall@3 is 0.125. Hotpot-to-QASPER and QASPER-to-Hotpot transfer both reach only 0.213. The adapter learns evidence alignment rather than a label-independent shortcut, but the geometry is not yet domain-general.

10.10 Recall as a Function of Selected Memory

A fixed k is a useful severe-sparsity test, but it changes meaning when documents contain different numbers of candidate chunks. We therefore reconstruct a second view at requested fractions $f \in \{0.01, 0.02, 0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.50, 1.0\}$, selecting $k_i(f) = \lceil fN_i \rceil$ separately for every example. Table 13 reports any-evidence recall because the historical artifacts retained the rank of the first annotated chunk, not the complete ordered evidence set. f_{80} and f_{90} are the first measured realized fractions reaching the corresponding recall after per-example ceiling, without interpolation. AUC_{0--30} is the normalized area over the sparse interval.

Table 13: Retrospective any-evidence recall versus requested parent-chunk fraction. The learned adapter averages five seeds and 160 seed–example evaluations. Other rows use historical 16- or 32-example cohorts and are not fully paired; differences across those rows are therefore descriptive. Per-example ceiling makes the realized fraction slightly larger than requested.

Mechanism	Dataset	R@5%	R@10%	R@20%	R@30%	f_{80}	f_{90}	AUC_{0--30}
Post-RoPE K mean	Combined	0.125	0.125	0.250	0.375	1.000	1.000	0.190
Pre-RoPE K mean	Combined	0.375	0.563	0.938	1.000	0.155	0.206	0.700
Centered-RoPE, $G = 8$	Combined	0.250	0.375	0.563	0.688	0.504	0.504	0.430
Hidden-state mean	Combined	0.438	0.625	0.625	0.813	0.306	0.504	0.574
Last-state query	Combined	0.531	0.719	0.875	0.906	0.158	0.306	0.724
Learned margin adapter	Combined	**0.675**	**0.831**	**0.956**	**1.000**	**0.105**	**0.158**	**0.835**
Learned margin adapter	HotpotQA	0.488	0.700	0.925	1.000	0.160	0.207	0.753
Learned margin adapter	QASPER	0.863	0.963	0.988	1.000	0.054	0.104	0.918

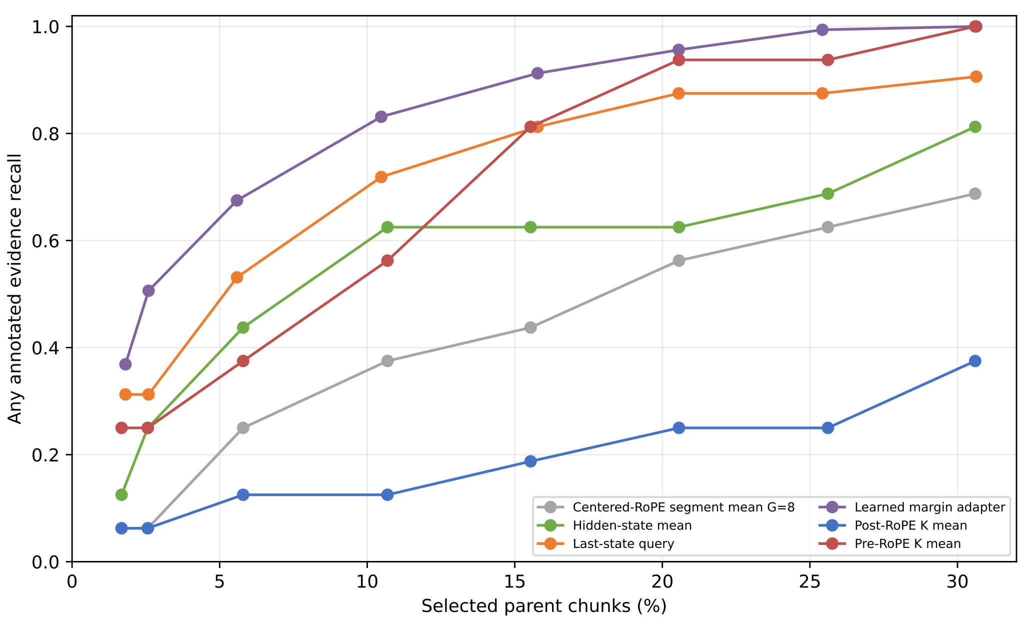

Figure 12: Any-evidence recall over the 0–30% selected-parent region. The learned metric has a clear 10–20% operating range that Recall@3 alone concealed. The strong pre-RoPE curve is from the earlier 16-example cohort and should not be read as a paired defeat of the learned adapter.

The fixed-k conclusion is therefore narrower than originally stated. The adapter does not meet the deliberately severe Recall@3 promotion gate, yet it retrieves some annotated evidence for 83.1%

of cases after selecting 10% of candidates and for 95.6% after selecting 20%. Even the parameter-free last-state query reaches 71.9% and 87.5% at those fractions. This is evidence for a practical sub-30% routing regime, not evidence of causal answer use. The HotpotQA–QASPER difference also remains material: QASPER reaches realized $f_{90} = 0.104$, whereas the multi-hop HotpotQA set requires 0.207.

For the frozen-feature test set, complete rankings and exact parent lengths permit a stricter multi-evidence calculation. Figure 13 plots the fraction of all annotated evidence chunks recovered against measured native-K/V tokens. At requested 10% and 20%, the selected seed recovers 0.356 and 0.442 of all annotated chunks, respectively; the corresponding any-evidence values are 0.844 and 0.906. These quantities answer different questions and must not be interchanged. Because nearly every parent has 32 tokens, the measured K/V fractions (0.104 and 0.205) closely track realized chunk fractions (0.105 and 0.206). We do not infer K/V fractions for older artifacts that lack exact token lengths.

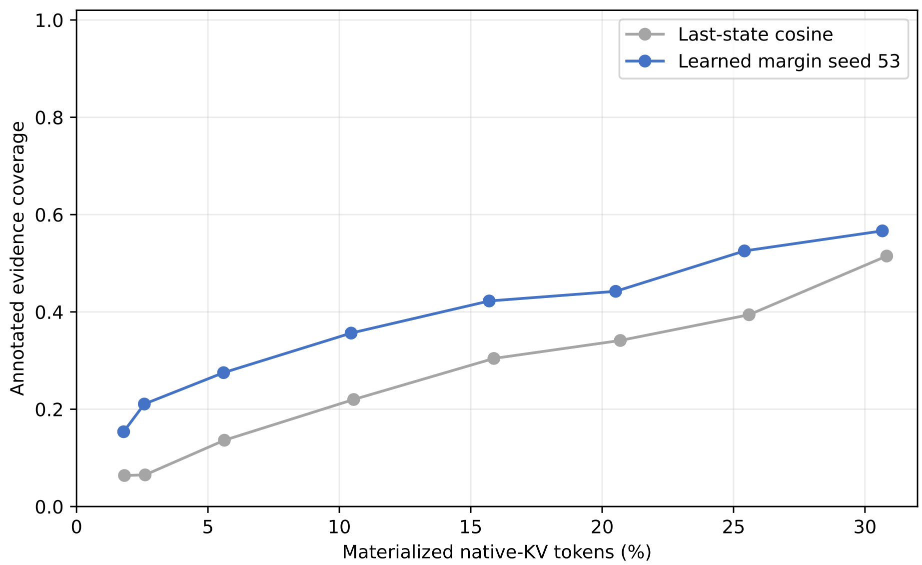

Figure 13: Exact annotated-evidence coverage versus measured materialized native-K/V fraction on the frozen-feature test set. This plot uses complete evidence sets and physical parent lengths; it is intentionally stricter than the any-evidence table.

The earlier 2×2 contrastive ablation also constrains the query interpretation. With cosine scoring, last/question-decay recall@3 is 0.469/0.250; with the learned metric it is 0.563/0.519. Learning removes much of the aggregate query's deficit, but aggregation adds no independent gain. The last state therefore remains the canonical query.

Increasing representation resolution is less effective than learning alignment. The selected single-mean adapter reaches recall@3 0.631 with a routing index equal to 0.392% of native detail K/V bytes. The earlier zero-shot token-resolution maximum reaches 0.375 while indexing 50%; it uses a smaller cohort and is therefore descriptive rather than paired. Width 256 doubles index cost to 0.784% without validation selection. Figure 14 shows the matched learned conditions. The raw 1024-dimensional float32 hidden-state index occupies 3.14% of detail K/V; the selected 128-dimensional index is eight times smaller.

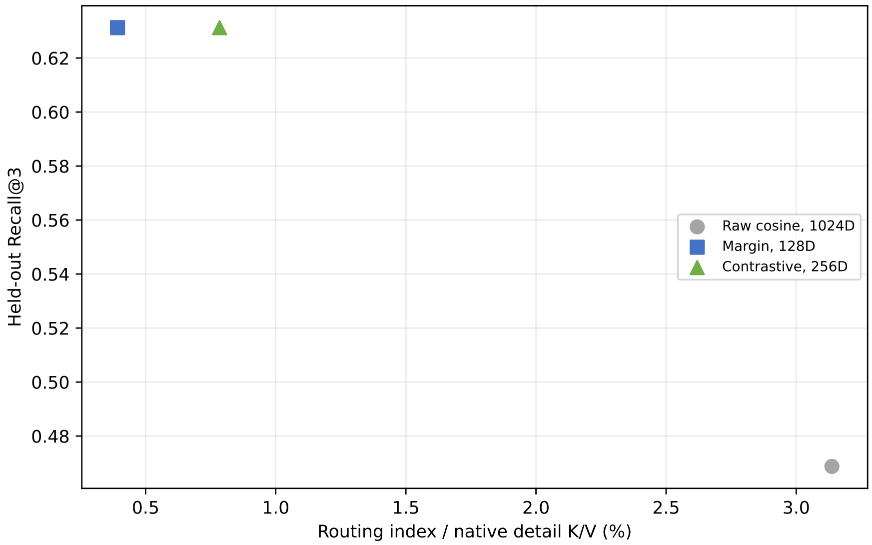

Figure 14: Held-out sparse recall versus routing-index storage for matched frozen-feature conditions. Width 128 improves the quality/cost point; width 256 adds cost without selection.

The integration changes routing only. At revision `a657bc7`, query pooling is implemented in `hf/query.py`, lines 77–117. Shared/asymmetric projections and cosine normalization are in `hf/routing_adapter.py`, lines 10–74. Reference publication projects only computed gists in `hf/injection.py`, lines 238–246, while the live query is projected in `hf/qwen.py`, lines 188–202. Native detail K/V bypass both projections. An invariant test confirms exact K/V equality with and without the adapter.

The implementation keeps this experiment auditable. At revision `604926e`, frozen feature publication sets every Qwen parameter non-trainable in `precompute_router_features.py`, lines 117–126. Deterministic mixed-negative selection is in `train_learned_router.py`, lines 171–239; contrastive and margin losses are at lines 246–283. These training paths consume stored vectors and cannot update native K/V.

Finally, the validation-selected seed is exercised through cache lookup, materialization, and generation on four held-out examples per dataset. It recalls evidence in 5/8 cases, covers 47.9% of annotated targets, stores a routing index equal to 0.393% of detail-K/V bytes, and incurs no native-limit violation. Frozen-feature adapter scoring takes 2.05 ms/example and full ranking 2.67 ms on the GTX 950M, versus 2.18 ms for raw cosine; these small synchronized timings are descriptive rather than a serving benchmark. Answer F1 is 0.090 with PRA disabled and 0.090 with learned routing enabled. Better selection does not establish causal use; further router capacity is not a justified response to that separate failure.

11 Does Correct Native Memory Change the Answer?

Retrieval recall answers whether a router found annotated evidence. It does not answer whether the frozen decoder can use that evidence after it becomes native K/V. The initial adapter consumed routed memory only at layer 27; the earlier oracle diagnostic activated at most layers 24–27. We therefore intervene at both the selection and consumption boundaries. The no-memory condition supplies only the question. Oracle conditions force every 32-token parent intersecting an annotated evidence span. The direct-text control places the same evidence in the ordinary prompt. Four deterministic examples from each dataset are used, so this remains a mechanism probe rather than an accuracy estimate.

Forced identities enter `prepare_selected_memory`, exactly where ordinary semantic routing ends. Both routes then share thresholding, whole-parent budgeting, native post-RoPE K/V, CPU-to-GPU transfer, GQA layout, source positions, additive masks, and Qwen's eager attention kernel. A selected parent identity is fixed across layers, but its tensor is not: layer ℓ receives $K_c^{(\ell)}, V_c^{(\ell)}$ captured from that layer's own Qwen projections and RoPE path. The 640-token bound reserves 128 direct tokens and permits up to 512 memory tokens per layer. Every oracle parent survives; no budget or native-limit violation occurs, and each reference-encoding call is at most 128 tokens. "Active K/V" is the sum of materialized physical tokens over consuming layers, not the number of distinct source tokens.

Generation-only F1/EM is too coarse for this small frozen checkpoint. The primary paired outcome is therefore teacher-forced mean gold-token log-probability,

$$\Delta_{c,i} = \frac{1}{|y_i|} \log p(y_i \mid q_i, c) - \frac{1}{|y_i|} \log p(y_i \mid q_i, \varnothing), \tag{14}$$

with first-token probability/rank, generated text, attention mass, and answer-position hidden state deltas retained in per-example traces. Four deterministic examples from each dataset are used. This is a mechanism probe, not an accuracy estimate.

Table 14: Oracle consumption-depth sweep. $\Delta \log p$ is paired mean gold-token log-probability relative to no memory. K/V is aggregate physical materialization across active layers. TTFT and generation are synchronized descriptive seconds, not serving benchmarks.

Dataset	Condition	Layers	$\Delta \log p$	F1	K/V tokens	TTFT	Generation
HotpotQA	none	0	.000	.056	0	.140	.737
HotpotQA	oracle, last 1	1	+.071	.056	112	.143	.733
HotpotQA	oracle, last 4	4	+1.004	.056	449	.154	.813
HotpotQA	oracle, last 8	8	+2.311	.056	898	.169	.877
HotpotQA	oracle, last half	14	+3.641	.056	1,572	.193	1.013
HotpotQA	oracle, all	28	-.766	.000	3,143	.245	1.382
HotpotQA	direct evidence text	0	+4.635	.125	0	.198	.986
QASPER	none	0	.000	.125	0	.135	.781
QASPER	oracle, last 1	1	-.091	.125	240	.137	.785
QASPER	oracle, last 4	4	-.161	.125	960	.149	.844
QASPER	oracle, last 8	8	+.347	.125	1,920	.163	.899
QASPER	oracle, last half	14	+.072	.100	3,360	.188	1.081
QASPER	oracle, all	28	-7.492	.000	6,720	.249	1.488
QASPER	direct evidence text	0	-1.791	.000	0	.416	1.090

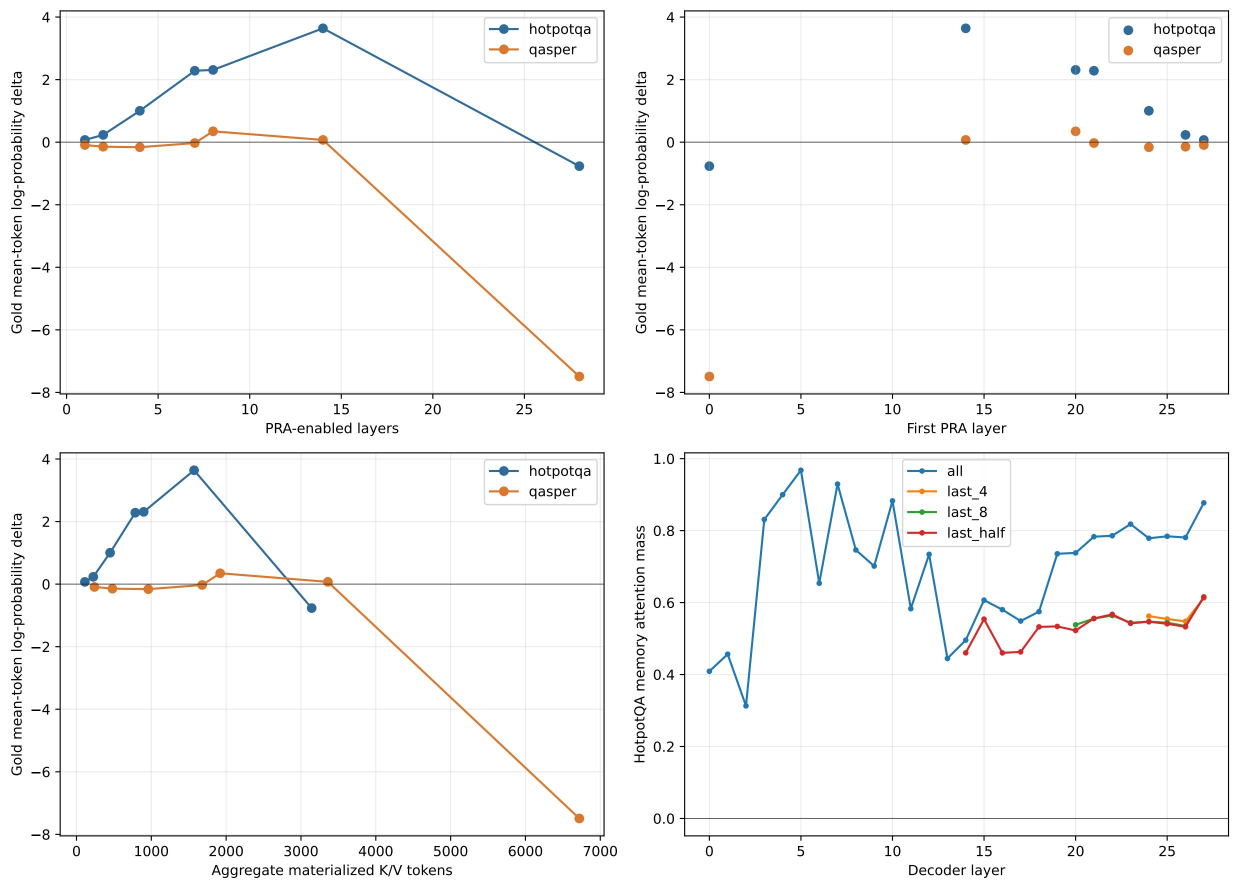

Figure 15: Oracle depth sweep. Repeated exposure through a contiguous late band improves HotpotQA likelihood, but the curve is non-monotone: all-layer replay is harmful on both datasets. The lower panels expose the associated aggregate materialization cost and per-layer attention rather than treating layer count as free.

HotpotQA supports the integration-depth hypothesis within a late band. The last-one, last-four, last-eight, and last-half interventions move mean log-probability by +0.071, +1.004, +2.311, and +3.641 nats/token. The last eight recover 49.9% of the direct-text effect while using 8/28 layers; the last half recover 78.6% at 1.75 times the last-eight materialization. Generated F1 does not change. The evidence therefore establishes causal likelihood use, not decoded QA success.

QASPER separates perturbation from interpretable utility. It turns positive at the last eight layers (+0.347) but is only +0.072 at the last half, while direct evidence text is negative (−1.791). That control failure prevents using this subset to grade PRA transport. Attention mass and hidden-state perturbations are retained in the artifact, but neither alone is evidence that a perturbation is useful.

11.1 Placement is not interchangeable

The depth result justifies one placement follow-up. Table 15 fixes either four or seven consuming layers and leaves oracle parent identities, positions, prompts, and per-layer payload sizes unchanged.

Table 15: Combined oracle placement result. Equal-count rows have equal aggregate K/V cost.

Placement	Layers	First layer	$\Delta \log p$	F1
Early contiguous	4	0	-10.573	.000
Middle contiguous	4	12	-.556	.090
Evenly spaced	4	0	-10.398	.000
Late contiguous	4	24	+.421	.090
Every fourth	7	0	-6.090	.000
Late contiguous quarter	7	21	+1.128	.090

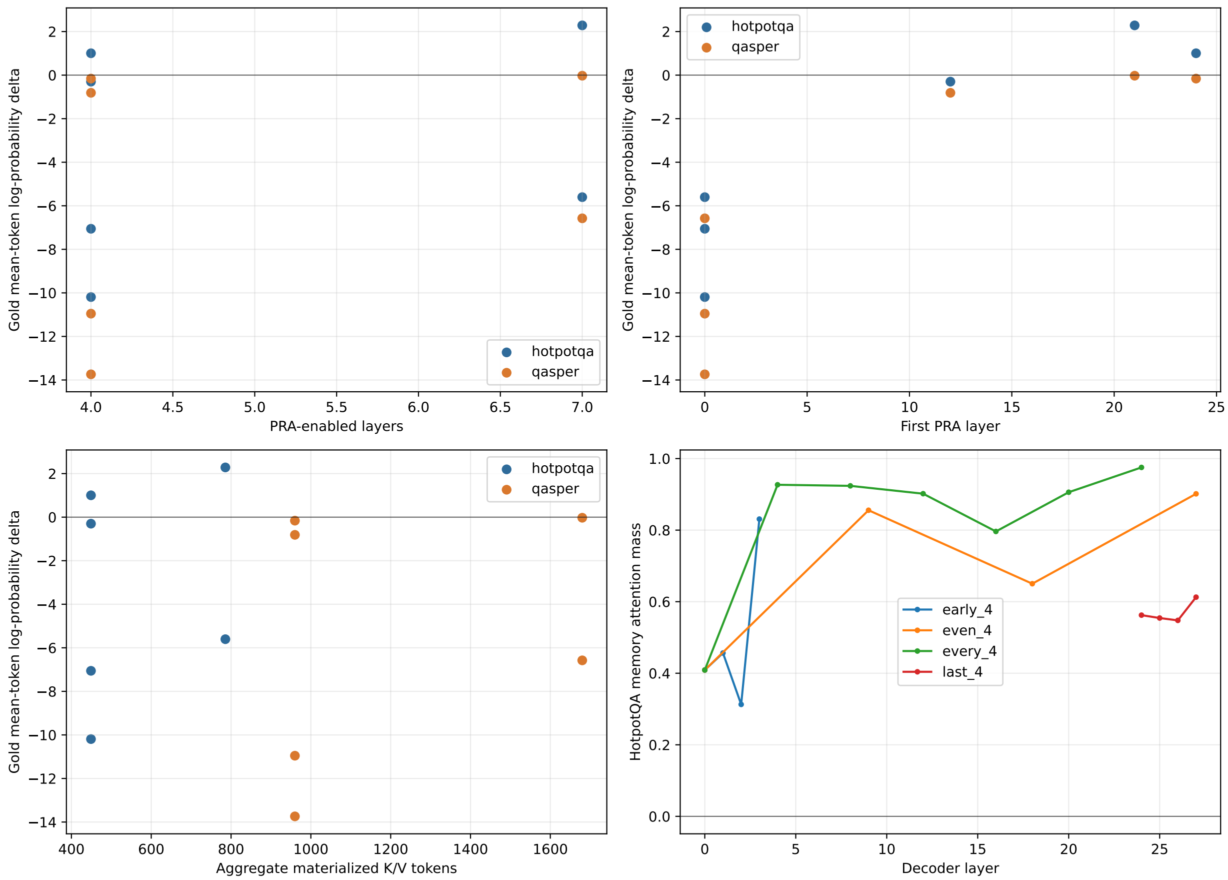

Figure 16: Oracle placement at matched parent identities and matched per-layer payload. Earlier and sparse placements are not approximations to a late contiguous band; they reverse the likelihood effect despite comparable attention and exactly equal K/V cost at equal counts.

This rejects both proposed shortcuts. Earlier insertion does not help by leaving more downstream depth, and every-fourth insertion does not approximate dense late consumption. The likely mechanism is a layer-specific contextualization/topology mismatch: independently encoded reference states are least compatible with early residual computation, whereas a contiguous upper band can repeatedly integrate them after the local question has become semantically formed. This interpretation is mechanistic, not yet a general theorem about decoder depth. All-layer replay is especially informative: more attention opportunities can make the answer worse. The operating point is therefore a band, not "as many PRA layers as possible."

11.2 Route once, consume with layer-native payloads

The routed follow-up scores the question once at layer 27, selects three parent IDs, and reuses those identities across either the last 8 or last 14 layers. Every layer resolves the ID to its own post-RoPE K/V; no tensor crosses layer boundaries. Table 16 reports the complete pre-top-k evidence-identity ranking and the aggregate consumption cost.

Table 16: Combined route-once result over eight examples. Recall is the fraction of mapped evidence-parent identities recovered at a parent-selection fraction; f_{80} and f_{90} are the first measured fractions reaching those recall levels.

Router	Consumption	R@5%	R@10%	R@20%	R@30%	f_{80}	f_{90}	$\Delta \log p$	F1	K/V
Learned margin	last 8	.264	.403	.521	.639	.502	1.000	+.227	.090	768
Learned margin	last half	.264	.403	.521	.639	.502	1.000	+.374	.028	1,344
Raw hidden-state cosine	last 8	.092	.224	.414	.573	1.000	1.000	+.150	.090	768
Raw hidden-state cosine	last half	.092	.224	.414	.573	1.000	1.000	+.568	.090	1,344

The learned router gives the stronger sparse ranking and reaches evidence in 5/8 selected top-3 sets. Last-eight consumption preserves the no-memory F1 and moves likelihood by +0.227; last-half consumption moves it by +0.374 but lowers F1 to .028. Raw cosine has weaker sparse recall yet a larger last-half likelihood shift. This is not a contradiction: a selected non-annotated parent can still change the answer distribution, and a likelihood movement need not cross the greedy decoding boundary. The learned last-eight condition is the conservative quality/cost operating point. It materializes 768 K/V tokens across layers, increases TTFT from .141 to .167 seconds and generation from .759 to .907 seconds, and adds one .149-second query encoding plus .021-second exact route. Its layer-27 learned index averages 45,184 bytes; storing projected gists for all 28 instrumented layers averages 1.27 MB, versus 322.7 MB of resident detail K/V. These eager single-example timings characterize this implementation only.

11.3 Canonical end-to-end confirmation

The earlier single-layer `#__head` run established bounded displacement but generated no correct answer. We repeat that probe with the learned router at layer 27 and the selected late-eight consumption band. The 416-token logical prompt displaces 288 tokens, retains a 128-token direct tail with positions 288–415, and indexes nine 32-token parents. The router ranks the evidence parent first and selects it with two neighboring parents. Each layer 20–27 then consumes its own 96-token native K/V payload.

Table 17: Multi-layer bounded-head confirmation. Rank is the gold first-token vocabulary rank; K/V is aggregate physical materialization over the consuming layers.

Condition	Evidence rank	$\Delta \log p$	First-token rank	K/V	F1	Viol.
No memory	–	.000	94,106	0	.000	0
Learned route, last 8	1	+5.469	1,351	768	.000	0

The intervention raises the two-token answer's mean log-probability from -10.924 to -5.455 and its first-token probability from 3.79×10^{-9} to 1.95×10^{-5}. Greedy output still omits "cobalt." The maximum native operation remains 128 tokens and no violation occurs. Thus the bounded head now demonstrates routing, multi-layer layer-native transport, and strong causal likelihood use together. It still does not establish decoded task success.

12 Systems Observations

The first run reaches 1,345,618,432 allocated GPU bytes, including the complete FP16 model and temporary activations. Full reference K/V remains on CPU; only selected detail K/V moves to the GPU. This is evidence that the residency path executes, not a memory-savings comparison: no matched dense long-context peak is measured here. Likewise, cold source construction, warm routing, generation, and transfer are reported separately so cache reuse is not hidden.

The artifacts keep three quantities distinct: compact routing-index bytes, resident detail-K/V bytes, and selected/materialized K/V bytes. The token-max experiment changes the first by roughly 32 times while holding the latter two fixed at a given parent selection and budget. Multi-layer consumption changes a fourth quantity: the same three parent identities correspond to 96 distinct physical K/V tokens per consuming layer. Last-eight and last-half route-once runs therefore materialize 768 and 1,344 physical tokens, respectively. They route once; they do not pay eight or fourteen semantic searches. The layer-27 learned routing index averages 45 KB, but this diagnostic implementation stores routing gists for every instrumented layer. A serving implementation that routes only at layer 27 can omit the other routing views while retaining the required per-layer detail K/V.

The current HF path is intentionally eager and unfused. Variable memory rows are padded into a rectangle before Qwen's native eager function. Flash attention, SDPA, custom kernels, ANN routing, asynchronous overlap, and serving concurrency remain outside this correctness phase.

13 PRA-HF Productization and Cross-Family Portability

The release implementation packages the validated mechanism as `pra-hf` rather than requiring callers to assemble experiment objects. Its public workflow is deliberately short:

```
from pra_hf import PRAConfig, PRAForCausalLM

model = PRAForCausalLM.from_pretrained(
    BASE_MODEL, routing_adapter=ROUTER,
    pra_config=PRAConfig(selected_fraction=0.20))
ref = model.add_reference_file("paper.txt")
result = model.generate(question, return_details=True)
model.remove_reference(ref)
```

The same object supports explicit URI/text pairs, in-memory text, `#__head` rollover, chat formatting, memory clearing, and request statistics. Configuration is JSON serializable. When both are present, `selected_fraction` takes precedence over `top_k`. The router first ranks every parent identity, the requested policy selects $\lceil fN \rceil$ parents, and the hard native-context budget may retain fewer whole chunks. The API therefore reports requested parent fraction, requested native-K/V token fraction, and actually materialized native-K/V fraction separately.

13.1 Stable router artifact

Each release router contains only `adapter_model.pt`, `config.json`, and a model card. Metadata pins the base model and revision, model family, routing representation and layer, 32-token parent size, dataset, split and training seeds, adapter architecture and dimension, metrics, PRA version, and source revision. It never republishes base-model weights. The selected asymmetric linear metric has

$$s(q, c) = \mathrm{norm}(W_q h_q)^\top \mathrm{norm}(W_c g_c). \tag{15}$$

Qwen uses 262,144 trainable router parameters, 0.044% of its frozen 596,049,920 parameters. SmolLM2 uses 147,456 parameters, 0.110% of its frozen 134,515,008 parameters. Both use a 128-dimensional FP32 routing vector, or 512 index bytes per parent. Reference K/V remains in the pretrained model's FP16 native-head layout.

13.2 Thin Llama-family adapter

SmolLM2-135M is a 30-layer Llama-family decoder with 9 query heads, 3 native K/V heads, and 64-dimensional heads. The Llama adapter does not copy the Qwen or Transformers forward method. Qwen and Llama in Transformers 4.55.4 expose the same projection, RoPE, cache-update, GQA, and output contracts, so the shared adapter path owns capture and PRA control flow. The 49-line Llama module only resolves Llama's installed `apply_rotary_pos_emb` and `eager_attention_forward`, then omits Qwen's sliding-window keyword. Injection dispatches by the concrete attention module.

On the real FP16 checkpoint, memory-disabled wrapping gives exact logits, every hidden state, generation-cache length, and greedy output. An enabled CUDA probe captures CPU-resident $[1, 3, 23, 64]$ post-RoPE K/V, transfers it without persistent head expansion, and produces finite memory-attended logits. Offline tiny-model tests independently cover exact disabled parity, native-head layout, active memory, and bounded long-prompt rollover.

13.3 Five-seed cross-family routing

For the portability release, both models use the same evidence-supervised protocol: frozen last-layer attention-input states, one mean per 32-token parent, asymmetric linear projection to 128 dimensions, 512 optimization steps, QASPER training, and seeds 11, 23, 37, 53, and 71. Selection uses QASPER validation AUC over the 0–30% region; HotpotQA is transfer stress rather than a selection set. Table 18 and Figure 17 report evidence-identity recall, which gives partial credit for multi-span evidence and reaches one only when all mapped evidence identities are recovered.

Table 18: Five-seed QASPER test recall–sparsity results. Values are mean ± sample standard deviation; f_{80} is the first measured requested fraction reaching 0.80 recall.

Family	R@5%	R@10%	R@20%	R@30%	$\mathrm{AUC}_{0:30}$	f_{80}
Qwen3-0.6B	0.293	0.362	0.454 ± 0.007	0.532	0.396	1.00
SmolLM2-135M (Llama)	0.243	0.334	0.447 ± 0.051	0.532	0.371	1.00

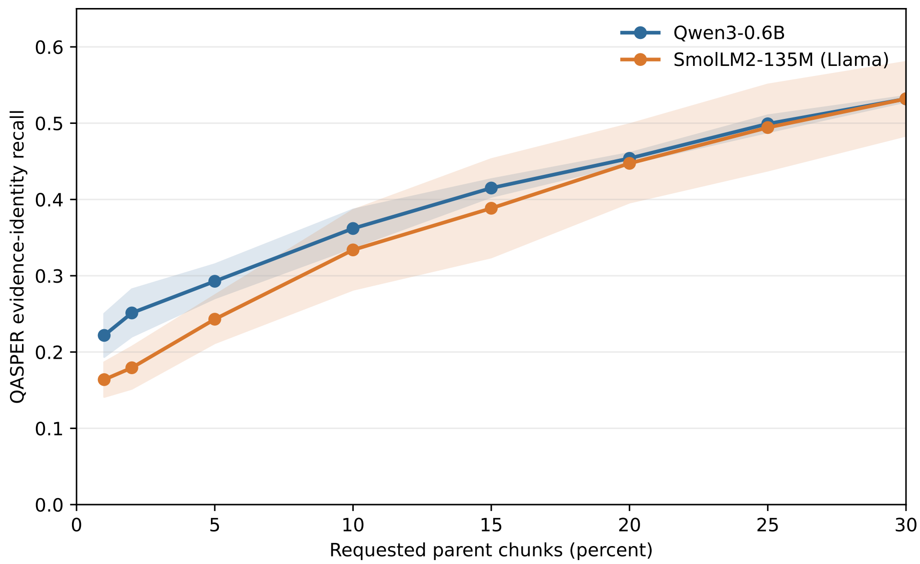

Figure 17: QASPER evidence recall versus requested parent fraction for five independently trained routers per family on the same 16-example test cohort. Bands show one sample standard deviation. Qwen is more seed-stable; neither family reaches high recall in the 10–20% operating region.

The selected Llama seed reaches R@20%=0.497 and $\mathrm{AUC}_{0:30} = 0.419$; the selected Qwen seed reaches 0.446 and 0.405. These selected values are useful artifact descriptors, but the five-seed matched-cohort means are the comparison. The mean R@20% values differ by only 0.007 and mean R@30% is 0.532 for both, but Qwen has the stronger sparse AUC and much lower R@20% seed variance. Llama's variance and both families' $f_{80} = 1.0$ prevent a claim of strong sparse retrieval. The result instead establishes that the same compact-router interface trains and executes across two native attention families.

13.4 Product-path systems and answer probes

We ran each packaged router through the public API on two held-out examples per dataset. This is a systems demonstration, not a statistically powered QA benchmark. Every request routes once and replays parent identities through the final eight layers under a 128-token materialization budget. Table 19 distinguishes the compact index, all CPU-resident layer-native detail K/V, and the subset transferred to each consuming layer.

Table 19: Four-example CUDA product-path means on a GTX 950M. Detail and index sizes vary with source length. Peak GPU includes the complete model. Routing is exact Torch/Python search.

Family	Evidence	F1	Requested	Materialized	Route ms	Peak GiB
Qwen3-0.6B	0.667	0.033	20.4%	6.61%	6.40	1.126
SmolLM2-135M	0.750	0.028	21.0%	6.25%	32.74	0.261

Qwen averages 45.9 KiB of routing index and 91.3 MiB of resident eight-layer detail K/V; SmolLM2 averages 48.8 KiB and 18.2 MiB. The 128-token cap explains why an approximately 20% router request becomes only 6–7% materialized K/V on these variable-length sources. Per request, the implementation transfers 4.0 MiB for Qwen and 0.75 MiB for SmolLM2 across the eight consuming layers. Mean routed wall time is 2.15 and 1.85 seconds for 16 generated tokens. There are no native-limit violations.

Retrieval and use remain separate gates. The four-example probe obtains substantial annotation coverage but lower routed F1 than the already weak no-reference baseline. This agrees with the larger Qwen causal study: correct memory can alter gold likelihood, but a frozen small decoder does not reliably turn that signal into decoded answers. The product release therefore exposes the measurements and retains this limitation rather than treating evidence recall as end-task success.

14 Limitations and Next Experiments

The mechanistic depth study uses one 596M-parameter Qwen checkpoint, while the portability study adds one smaller Llama-family checkpoint. The representation/chunk and exploratory multi-gist sweeps use 16 matched examples; confirmation uses deterministic subsets rather than a full validation corpus, and four examples overlap between seeds. The oracle layer-depth study has only four examples per dataset, reuses one fixed condition order, and is a mechanism probe rather than an accuracy or serving estimate. Its HotpotQA direction must be replicated at larger scale; its QASPER condition fails the direct-text upper control. The main sweeps measure annotated-span ranking, while generation remains too weak for an end-task claim. There is no matched learned text retriever, LoRA adaptation, serving-throughput study, or full dense-context quality comparison.

The tiny learned adapter establishes that evidence supervision can reshape frozen hidden states, especially on QASPER, but validation/test instability and weak cross-domain transfer prevent a general routing claim. Fraction-based evaluation shows that it has a useful 10–20% retrieval operating region despite failing Recall@3; this revises the gate interpretation, not the causal generation result. Multi-layer oracle and bounded-head results show that frozen Qwen can use selected native K/V without a learned memory-use adapter; such an adapter is therefore not the next justified intervention. The next discriminating experiment should replicate the late-band result on a larger, checkpoint-solvable cohort, then jointly report decoded quality and likelihood at the last-eight operating point. Routing data should also increase in split size and label diversity before adding adapter capacity. The contiguous-mean, token-max, centered-RoPE, and query-pooling results close the current simple zero-parameter sweeps. The Llama port establishes software and tensor-contract portability, not transfer of Qwen routing quality or solved QA. Gemma, serving kernels, and larger checkpoints remain later work.

15 Related Work

PRA differs from text retrieval-augmented generation, which retrieves text and lets the model encode it in the current prompt [5]. It retrieves layer-native K/V after a separate bounded encoding path. This makes PRA complementary to RAG: a text retriever may choose documents before PRA routes chunks inside cached model state. Sparse-attention systems such as Longformer and BigBird constrain token-token connectivity inside a forward pass [2, 11]; PRA instead materializes selected state from an external addressable cache. KV-cache systems emphasize serving capacity and token retention policies [4, 12, 6]. Paper 2 asks whether selection can be inserted beneath an existing pretrained family without changing its disabled behavior.

16 Conclusion

One shared PRA core now serves the controlled model plus thin Qwen and Llama-family adapters. Both real checkpoints pass exact disabled-path gates; active adapters preserve native RoPE, GQA cache layout, layer-native K/V, masks, cache growth, and output projection. They replay one selection through a contiguous late band, keep full references on CPU, and execute implicit long-prompt heads under a hard native-operation bound. The routing study adds a second architectural result: semantic routing state and native attention state are separate interfaces. Attention-input hidden-state gists form the canonical compact index, while a fixed selected identity always materializes the same native post-RoPE K and native V. A direct invariant test establishes identical K/V and attention output for router and oracle labels that provide the same set.

That correction is necessary but insufficient. Hidden-state routing reproduces useful evidence ranking above the post-RoPE baseline, yet sparse recall and MRR remain below the declared gate, especially on QASPER. Selecting nearly one quarter of chunks raises recall@16 to 0.797, but does not constitute a sparse-routing solution. Contiguous multi-gist means increase index size and occasionally improve broad recall without improving recall@3 or MRR. Token-resolution maxima raise some HotpotQA ranks but worsen sparse QASPER recall at 50% routing-index overhead. Centered-RoPE gists repair part of incompatible-frame averaging and, with finer segments, closely reproduce native token-QK ordering. They still trail pre-RoPE and hidden-state routing on annotated evidence. This directly supports the routing/payload split: native attention compatibility is not semantic evidence relevance. Recent-state and question-span query aggregation also fail to beat the last state. A tiny margin-trained metric improves combined recall@3 by 0.163 and balances HotpotQA/QASPER without positional bias, showing that relevance geometry is a real part of the bottleneck. Fraction-based analysis further shows any-evidence recall of 0.831 at 10% and 0.956 at 20%, so Recall@3 understates the usable retrieval regime. More dimensions and harder sampled negatives do not improve it. The validation-selected adapter still misses the held-out fixed-k gate and transfers poorly. Generated answer quality remains unchanged even when evidence is selected, but the expanded causal diagnosis revises the earlier conclusion. On HotpotQA, repeated oracle consumption in a contiguous late band produces a monotone gain through the last half and recovers most of the direct-text likelihood effect. Early, sparse, and all-layer placement are harmful, so layer count alone is not the mechanism. Route-once learned retrieval through the last eight layers produces a smaller positive likelihood change at 4.9% selected parents while preserving F1. The bounded `#_head` probe produces a large gold-likelihood gain when its rank-1 fact is consumed through that band. Frozen Qwen therefore can causally use retrieved native K/V; a learned memory-use adapter is not yet required. It cannot yet decode the answer reliably, and QASPER fails its direct-text control. The next work must replicate the late-band operating point at scale and separate retrieval generalization, checkpoint/task competence, and the remaining likelihood-to-decoding gap before claiming solved long-context QA. The productization study adds a narrower positive result: the family boundary, stable router artifact, reference API, fraction policy, and bounded execution transfer to Llama without copying the PRA mechanism. Across five QASPER-trained seeds on matched test identities, Qwen and SmolLM2 have similar mean recall at 20–30% requested chunks, but Qwen is substantially more stable and has the stronger sparse AUC. Neither reaches 0.80 evidence recall before exhaustive selection. The released API and checkpoints therefore make the present capability reproducible and useful for further evaluation, but they do not justify a long-context QA quality claim.

A Code-Level Adapter Reimplementation Guide

This appendix gives the minimum code structure needed to reproduce the integration without copying the Qwen implementation. Source references use physical line numbers at repository commit `db50846`. This checkpoint makes attention-input hidden-state routing the explicit default and exposes the fixed-selection materialization boundary. The line references are deliberately revision-pinned because upstream Transformers APIs and local diagnostics will move over time. The multi-layer identity mapping, experiments, tests, and complete artifacts are pinned separately at commit `8f6ef41`; all multi-layer line references below refer to that revision.

A.1 Tensor and ownership contract

A new family should add projection, position, cache, mask, kernel, and output conventions in its adapter, while leaving selection and memory policy in the shared core.

The canonical tensor contract is

$$Q \in \mathbb{R}^{B \times H_q \times T_q \times d_h}, \qquad K, V \in \mathbb{R}^{B \times H_{kv} \times T_k \times d_h}. \tag{16}$$

Reference entries use $B = 1$ and keep H_{kv}, including for GQA or MQA. They are not expanded to H_q in persistent storage. The controlled replay implementation shows the only required expansion point in `memory_batching.py`, lines 216–290: after memory and local K/V are joined, native K/V heads may be repeated for the attention computation. Qwen's installed eager kernel performs the corresponding operation in the production adapter.

A.2 Minimal forward-path skeleton

The following pseudocode is the adapter's essential control flow. It is intentionally family-neutral; "native" operations must call the installed model implementation rather than reimplementing its mathematics.

```
forward(hidden, native_positions, native_mask, past):
    if capture_is_armed:
        q_pre, k_pre, v = native_project_and_norm(hidden)
        q_post, k_post = native_position(q_pre, k_pre)
        save_transient_routing_features(hidden, q_pre, q_post, k_pre)
        save_post_position_detail_kv(k_post, v)

    if memory_is_disabled or reference_cache_is_empty:
        return original_attention(hidden, native_positions,
                                  native_mask, past)

    q_pre, k_pre, v = native_project_and_norm(hidden)
    q_post, k_post = native_position(q_pre, k_pre)
    if past is not None:
        k_post, v = past.update(k_post, v, layer_id, native_kwargs)
    route_q = matched_query(hidden[:, -1, :], q_pre, q_post)
    memory = shared_core.prepare_memory(q_post, route_q,
                                        direct_tokens=k_post.shape[2])
    if memory.is_empty:
```

```
20         return native_attention_and_output(q_post, k_post, v, native_mask)
21     k, v, mask = shared_core.combine_local_and_memory_kv(
22         k_post, v, memory, native_mask, query_tokens=q_post.shape[2])
23     return native_attention_and_output(q_post, k, v, mask)
```

Lines 8–10 are a correctness boundary, not an optimization. The Qwen implementation delegates to the untouched module at `hf/qwen.py`, lines 190–201, which makes disabled behavior auditable and bit-exact. In the active path, projection and RoPE occur at lines 204–207 and `Cache.update` occurs once at lines 208–215. Lines 216–221 choose the matched routing query without altering the post-RoPE attention query. If routing returns no materialized memory, lines 224–234 call the native kernel directly. Delegating at that point would execute a second cache update and duplicate generation history.

Memory is composed only after the native cache update. The shared operation at `core.py`, lines 394–436, pads variable memory rows, prepends memory K/V, creates an all-visible additive mask for valid memory positions, masks padding, and concatenates the unchanged local mask. The adapter then invokes Qwen's eager function and one pretrained `o_proj` at `hf/qwen.py`, lines 92–110 and 246–258. Thus local and memory positions share one softmax and one output projection.

A.3 Optional learned routing metric

The learned adapter is a replaceable routing component, not part of native attention. For an asymmetric linear metric,

$$r_q = \mathrm{norm}(W_q h_q), \qquad r_c = \mathrm{norm}(W_c g_c), \qquad s(q,c) = r_q^\top r_c. \tag{17}$$

A shared adapter aliases $W_q = W_c$. An independent implementation needs only the following boundary, revision-pinned at commit `a657bc7`:

```
# Reference publication, after ordinary hidden-state gist pooling
routing_gist = adapter.project_memory(routing_gist)
cache.store(routing_gist, native_post_rope_k, native_v)

# Live routing; native attention continues to use q_post
information_need = aggregate_query_states(h_in, strategy)
routing_query = adapter.project_query(information_need)
selected_ids = cache.search(routing_query)
k_mem, v_mem = cache.materialize_native_kv(selected_ids)
native_attention(q_post, concat(k_mem, k_local),
                 concat(v_mem, v_local))
```

The projection implementation is `hf/routing_adapter.py`, lines 10–74; checkpoint loading and freezing are lines 77–94. Memory projection occurs at `hf/injection.py`, lines 238–246, and live query projection at `hf/qwen.py`, lines 188–202. Cast hidden features to the adapter parameter dtype at this boundary; the checkpoint is FP32 while Qwen emits FP16 states. Do not cast, project, or recompute native detail K/V. Their exact-equality invariant is the test that prevents a routing retrofit from silently becoming an attention rewrite.

A.4 Oracle selection and causal diagnostics

An oracle must replace only identity selection. Adding a separate K/V injection path would confound semantic correctness with different budgeting, positions, masks, or attention code. The operational handle activates a layer set and assigns optional row-local identities in `hf/injection.py`, lines 69–89. The Qwen adapter chooses ordinary routing or the fixed set at `hf/qwen.py`, lines 273–286; both enter `core.py::prepare_selected_memory`, lines 337–388. The latter still applies the trigger, deduplication, whole-chunk budget, overlap policy, residency transfer, and rectangular packing. The fixed-set setter and its ownership contract are in `hf/adapter_base.py`, lines 71–83.

The experiment constructs oracle identities by intersecting annotation token spans with parent logical spans in `run_oracle_memory_use.py`, lines 107–134. Lines 194–281 append the gold continuation, gather its autoregressive token probabilities, retain opt-in eager-attention weights, and use temporary decoder-layer hooks for answer-position hidden states. Attention capture is disabled immediately afterward and is never part of ordinary inference. Conditions and paired deltas are assembled at lines 310–419. A reimplementation should assert three invariants per example: requested oracle IDs equal retained IDs, budget rejection is zero, and the largest native operation stays below the declared bound.

A.5 Route-once multi-layer consumption

Multi-layer PRA shares an identity decision, not an attention tensor. An implementation should make the distinction visible in its types:

```
selected = route_once(query_at_layer_27)       # parent IDs + scores
for layer in consumption_layers:
    layer_hits = []
    for hit in selected:
        chunk = cache[hit.uri].layer[layer].by_id[hit.chunk_id]
        assert chunk.kv.was_projected_by == layer
        layer_hits.append(hit.with_payload(chunk.kv))
    adapter[layer].set_fixed_selection(layer_hits)
model(question)                                # no further semantic search
```

The operational API activates an arbitrary layer set at `hf/injection.py`, lines 70–93. Identity-to-payload resolution is lines 97–145: for each requested layer it looks up the stable `chunk_id` in that layer's `LayerReferenceMemory`, replaces the payload, and records the source routing layer. A missing layer or chunk is an error rather than a fallback to another layer's K/V. The Qwen adapter's fixed-selection branch at `hf/qwen.py`, lines 273–289 then calls the same budgeting, transfer, mask, and attention path as ordinary routing.

The experiment captures the last-layer information-need state and performs one complete ranking at `run_multilayer_pra.py`, lines 161–198. Oracle and routed identities are mapped at lines 277–325; the remainder of the function records per-layer attention, hidden-state changes, materialized tokens, transfers, and latency. The invariant test at `tests/test_hf_integration.py`, lines 548–586 asserts equal parent IDs but different target-layer objects, tensor pointers, and values while preserving native $[B, H_{kv}, T, d_h]$ shape. This catches the most dangerous superficially working implementation: copying $K^{(27)}, V^{(27)}$ into every consuming layer.

A.6 Publishing a reference

Reference construction is a bounded prefill, not a second attention mechanism. A portable implementation follows this recipe:

```
disable_PRA_memory_during_reference_encoding()
for [block_start:block_end] in bounded_encoding_blocks(source):
    position_ids = logical_source_positions(block_start, block_end)
    arm_post_position_kv_capture(position_ids)
    model(block_ids, position_ids=position_ids, use_cache=False)
    captured = consume_native_kv_for_each_PRA_layer()
    for routing_window in overlapping_windows(captured):
        gist = pool(routing_window.attention_input_hidden_state)
        publish(uri, logical_offsets, post_rope_k, native_v, gist)
        discard_token_level_hidden_state()
restore_previous_memory_state()
```

The concrete loop is `hf/injection.py`, lines 235–354. Encoding block size limits one native model invocation; routing chunk size and overlap independently determine index granularity (lines 266–292). Each chunk slices the captured post-RoPE K/V, chooses transient routing features through `_routing_points` (lines 172–233), computes one or more compact gists, and records logical source offsets (lines 293–348). Keeping those two scales separate is important: changing index granularity must not silently change how the pretrained model encoded the source.

The contiguous multi-gist extension keeps that ownership boundary explicit. For a requested count G, split the routing feature rows into balanced source-order ranges, mean each range, score all resulting $[G, D]$ rows, reduce to one parent score, and preserve the winning local index for diagnostics. Top-k must then operate on parent identities before budgeting; otherwise several winning gists could duplicate one parent K/V payload. The reference implementation is revision-pinned at `0205df8`: pooling is in `gists/segment_mean.py`, lines 11–56, publication remains `hf/injection.py`, lines 163–190, and packed scoring plus max reduction remains `memory.py`, lines 487–593.

The same operation implements displaced long-prompt history. At `hf/injection.py`, lines 357–370, the prefix is published under the reserved `#__head` URI and the direct tail retains continued logical position IDs. Every reference-encoding call and every direct call must satisfy the model-safe native bound even when the logical source is longer.

A.7 Porting another decoder family

A practical port begins by locating seven hooks in the installed family: the attention module inside each decoder layer; Q/K/V projections; any Q/K normalization; positional encoding; generation-cache update and its keyword arguments; native mask convention; and the eager attention plus output projection. Implement the six abstract methods declared at `hf/adapter_base.py`, lines 85–106, then add a narrow type-selection branch analogous to `inject_pra` at `hf/injection.py`, lines 379–428. No checkpoint conversion or parameter copy should be needed.

The recommended validation order is:

1. With memory disabled, require exact logits, hidden states, greedy tokens, and generation- cache lengths against an unwrapped copy (test lines 55–72).

2. Capture a short reference and verify post-position state, logical offsets, device residency, and H_{kv} rather than H_q storage (lines 74–93).
3. Verify matched pre/post capture, representation switching, unchanged post-RoPE detail K/V, GQA grouping, serialization, and device parity (lines 109–214).
4. Supply one selected identity through router and oracle traces; require identical native K/V and attention output (lines 241–280).
5. Compare GQA replay with explicit head repetition on a synthetic tensor (lines 216–239).
6. Exercise an over-limit logical prompt and verify bounded native operations plus continued position IDs (lines 282–313).
7. Generate with active memory and assert that direct cache length grows by exactly one per decoded token (lines 315–330).

Four mistakes deserve explicit tests. Applying position encoding again to stored post-position keys changes their basis. Permanently expanding GQA K/V multiplies memory without adding information. Replacing the family mask with a generic causal mask can alter padding or sliding-window behavior. Finally, calling the wrapped module after manually updating its cache appends the same local token twice. Passing transport and parity tests establishes a faithful adapter; it does not establish that the router selects causally useful evidence, which requires the separate quality controls reported in the main paper.

A.8 Llama port and product API

The release implementation is pinned at commit `809694b`. The common abstract family contract remains in `adapter_base.py`, lines 32–132. Qwen's implementation resolves its installed helper functions through an overridable method at `qwen.py`, lines 38–77. The Llama module at `llama.py`, lines 8–48, subclasses that shared projection/capture/control path, replaces the symbol resolver, and invokes Llama's native eager function without the Qwen sliding-window argument. This is inheritance of a family-neutral tensor contract, not reuse of Qwen parameters. Injection selects Qwen or Llama from the concrete attention module at `injection.py`, lines 380–430.

An independent family port should make the same decision only after comparing installed forward signatures and testing every shared assumption. For Llama those assumptions are: Q/K/V linear outputs reshape to $[B, H, T, d_h]$; RoPE receives the same cosine/sine tuple; cache update accepts the layer index, sine, cosine, and cache position; the eager kernel repeats native K/V heads internally; and output returns token-major states before one native output projection. If any family differs, override that specific abstract method rather than adding a family branch to the router, cache, or budgeter. Tests in `test_hf_llama_integration.py`, lines 51–106, cover exact disabled behavior, native-head reference payloads, active memory, and rollover.

The public layer composes rather than replaces those internals. `pra_hf/config.py`, lines 14–127, validates stable product names, resolves negative layer indices, and translates them to the core configuration. `pra_hf/router.py`, lines 14–110, defines the versioned weights/metadata/model-card artifact and legacy-checkpoint conversion. The high-level model in `pra_hf/model.py`, lines 42–358, owns tokenizer/model loading, router compatibility, reference lifecycle, fraction selection, route-once identity replay, generation, chat, and memory economics. In particular, lines 192–258 flatten complete reference-grouped rankings, apply a global fraction or top-k cutoff, reconstruct typed identities, and map those IDs to each consuming layer's native payload. The CLI in `pra_hf/cli.py`, lines 22–145, is a thin caller of the same API; it does not maintain a second execution path. Product tests at `test_pra_hf_api.py`, lines 66–132, pin config serialization and precedence, router save/load and freezing, add/remove/clear, fraction routing, generation, statistics, and the rule that a router cannot change after gists have been indexed.

A.9 Revision-pinned source map

Table 20 connects the preceding control flow to the implementation.

Table 20: Source map for an independent adapter implementation.

Responsibility	File and physical lines
Abstract family boundary	`src/pra_torch/hf/adapter_base.py`, 16–106
Qwen projections, RoPE, native kernel, and output	`src/pra_torch/hf/qwen.py`, 53–110
Llama native-symbol binding and eager invocation	`src/pra_torch/hf/llama.py`, 8–48
Matched routing/detail capture	`src/pra_torch/hf/qwen.py`, 109–148; `src/pra_torch/hf/injection.py`, 80–210
Disabled and active runtime paths	`src/pra_torch/hf/qwen.py`, 171–275
Routing query and GQA reduction	`src/pra_torch/core.py`, 92–121
Budget, deduplication, transfer, and materialization	`src/pra_torch/core.py`, 155–392
Fixed-selection identity/payload boundary	`src/pra_torch/core.py`, 335–392
Route-once layer-native identity mapping	`src/pra_torch/hf/injection.py`, 70–145
K/V and additive-mask composition	`src/pra_torch/core.py`, 394–436
Reference publication and selected-layer injection	`src/pra_torch/hf/injection.py`, 235–428
Depth schedules and causal protocol	`experiments/paper2_hf/qa/run_multilayer_pra.py`, 62–964
Parity, routing, GQA, head, cache, and selection gates	`tests/test_hf_integration.py`, 80–636
Stable config, router artifact, and public API	`src/pra_hf/config.py`, 14–127; `src/pra_hf/router.py`, 14–110; `src/pra_hf/model.py`, 42–358
Llama and product-facing gates	`tests/test_hf_llama_integration.py`, 51–106; `tests/test_pra_hf_api.py`, 66–132

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